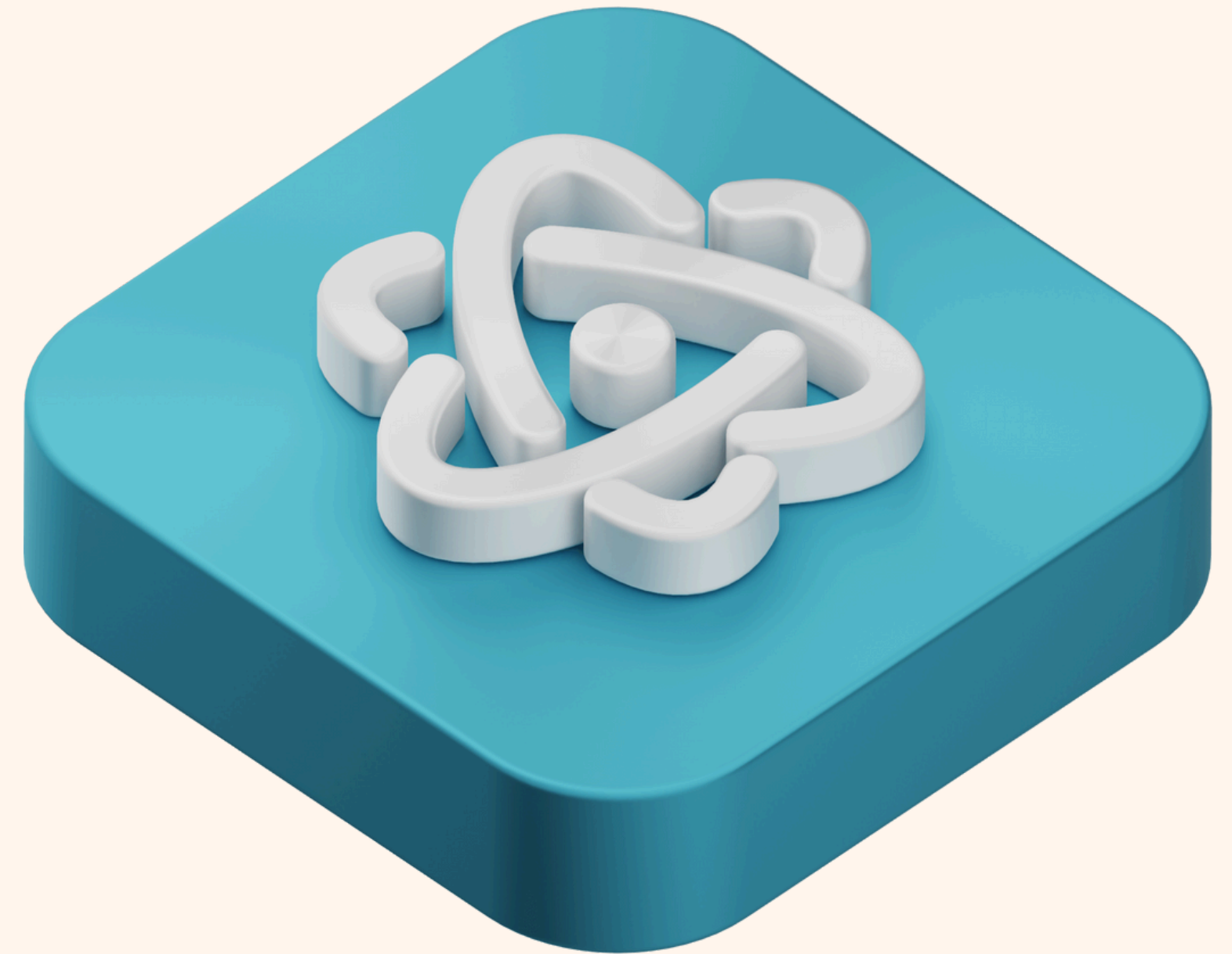
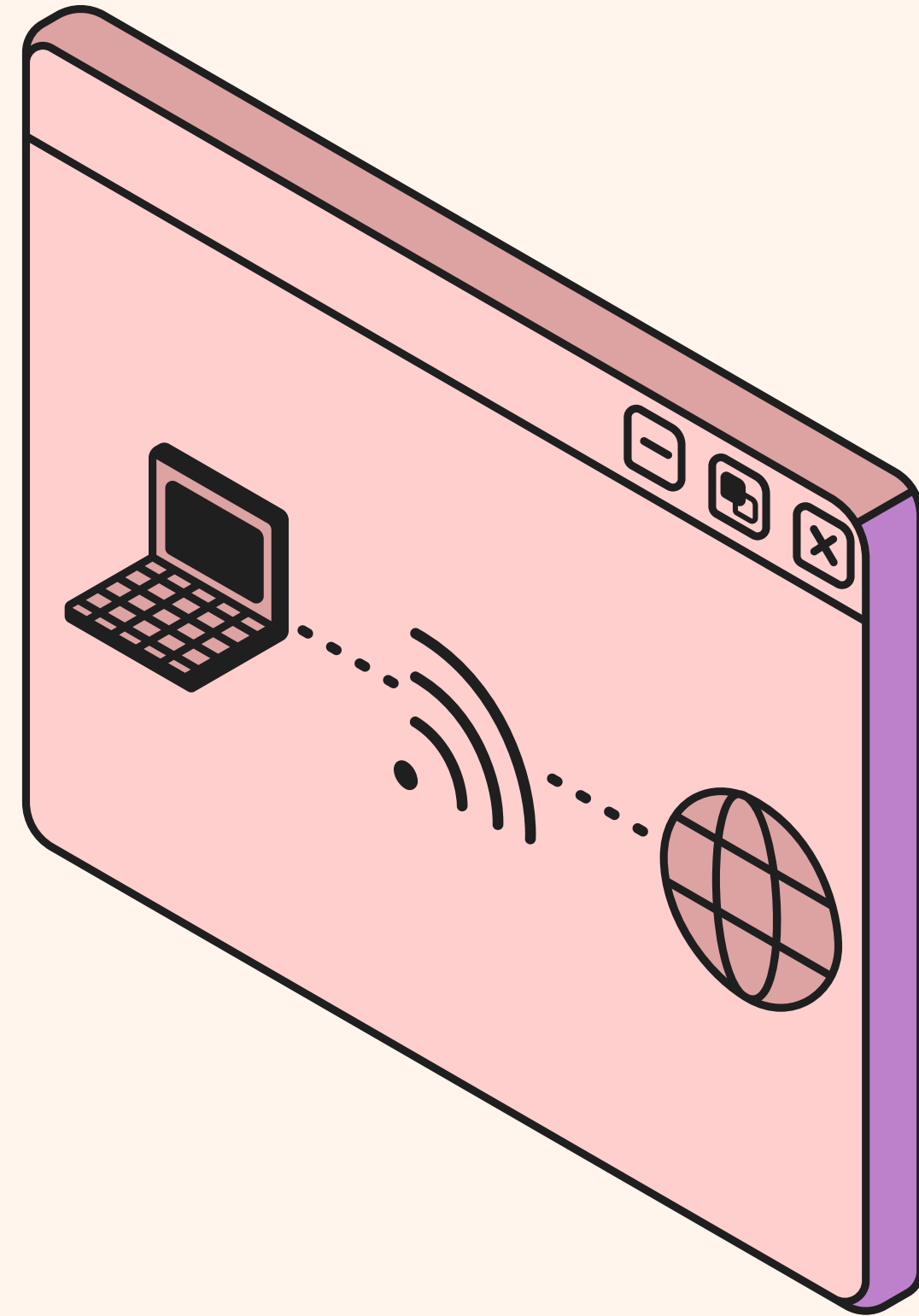


FRONTEND DEEP DIVE



AGENDA

1. How to learn + classroom culture
2. How browsers work



META-COGNITION

**Answer the following in your
learning doc:**

1. How do you learn best?

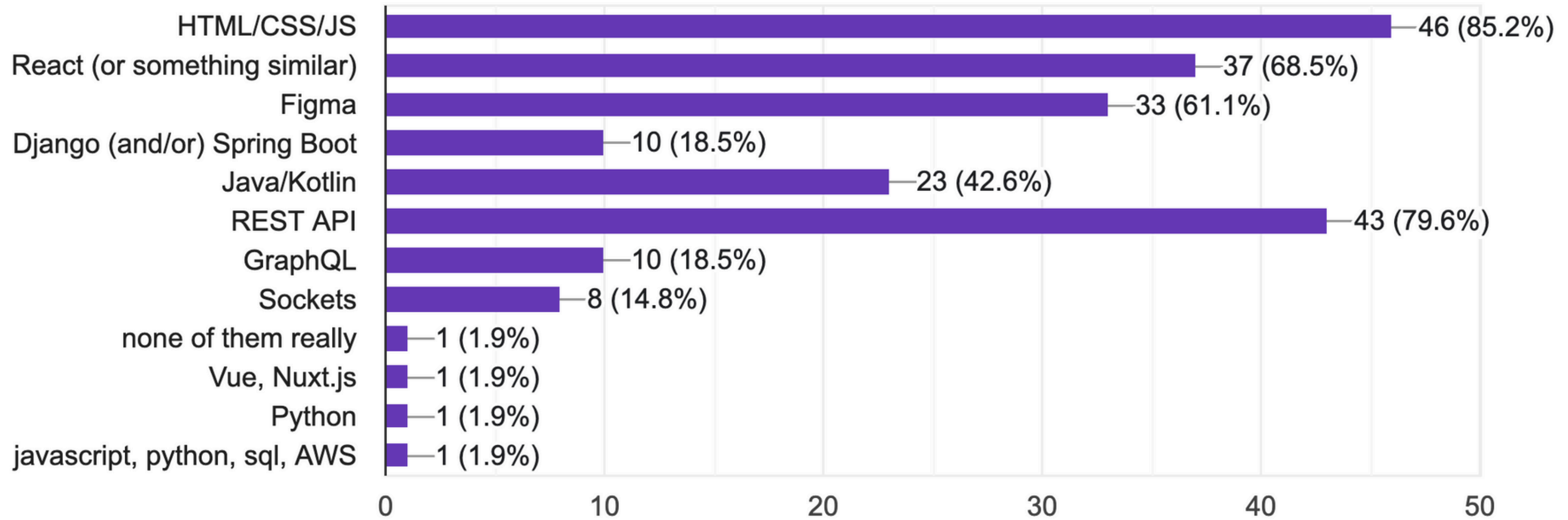
No doc? Instructions here: [bitly/42dwJqO](https://bit.ly/42dwJqO)

CLASSROOM CULTURE

bit.ly/cs303culture

Which of the following have you previously worked with?

54 responses

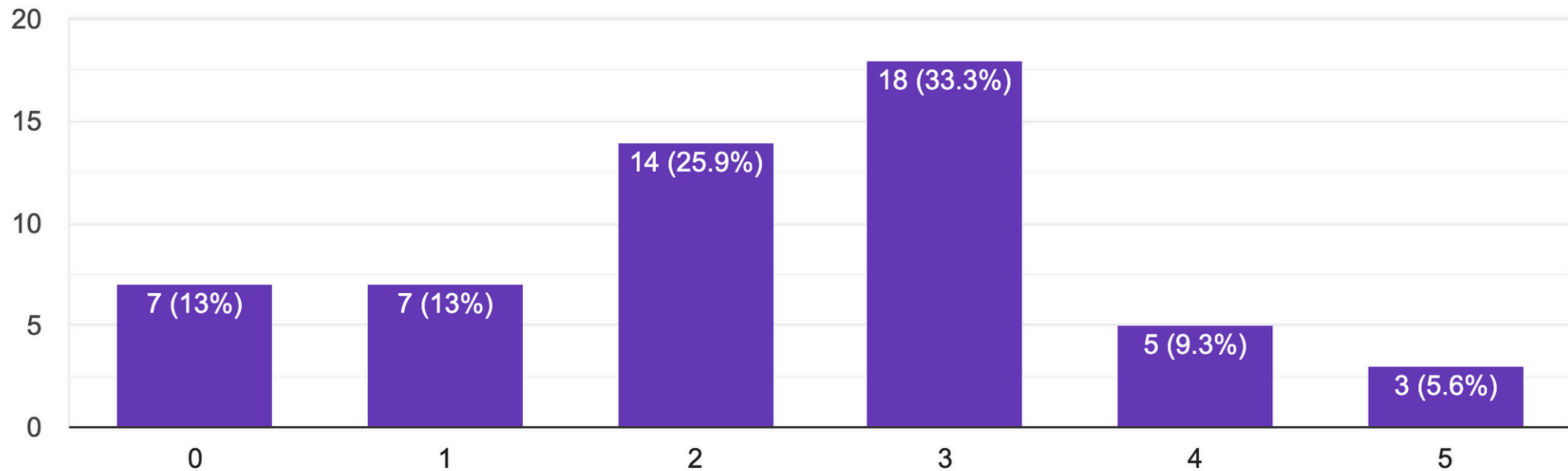


EGO IS THE ENEMY...

COLLABORATION IS KEY

How confident are you in building full stack apps from scratch? Ex: Building both the frontend and backend, then connecting the two

54 responses



HOW BROWSERS WORK

Why do we need to know this?

- Allows us to understand what exactly is happening when we create a React component
- Latency matters and it impacts the UX

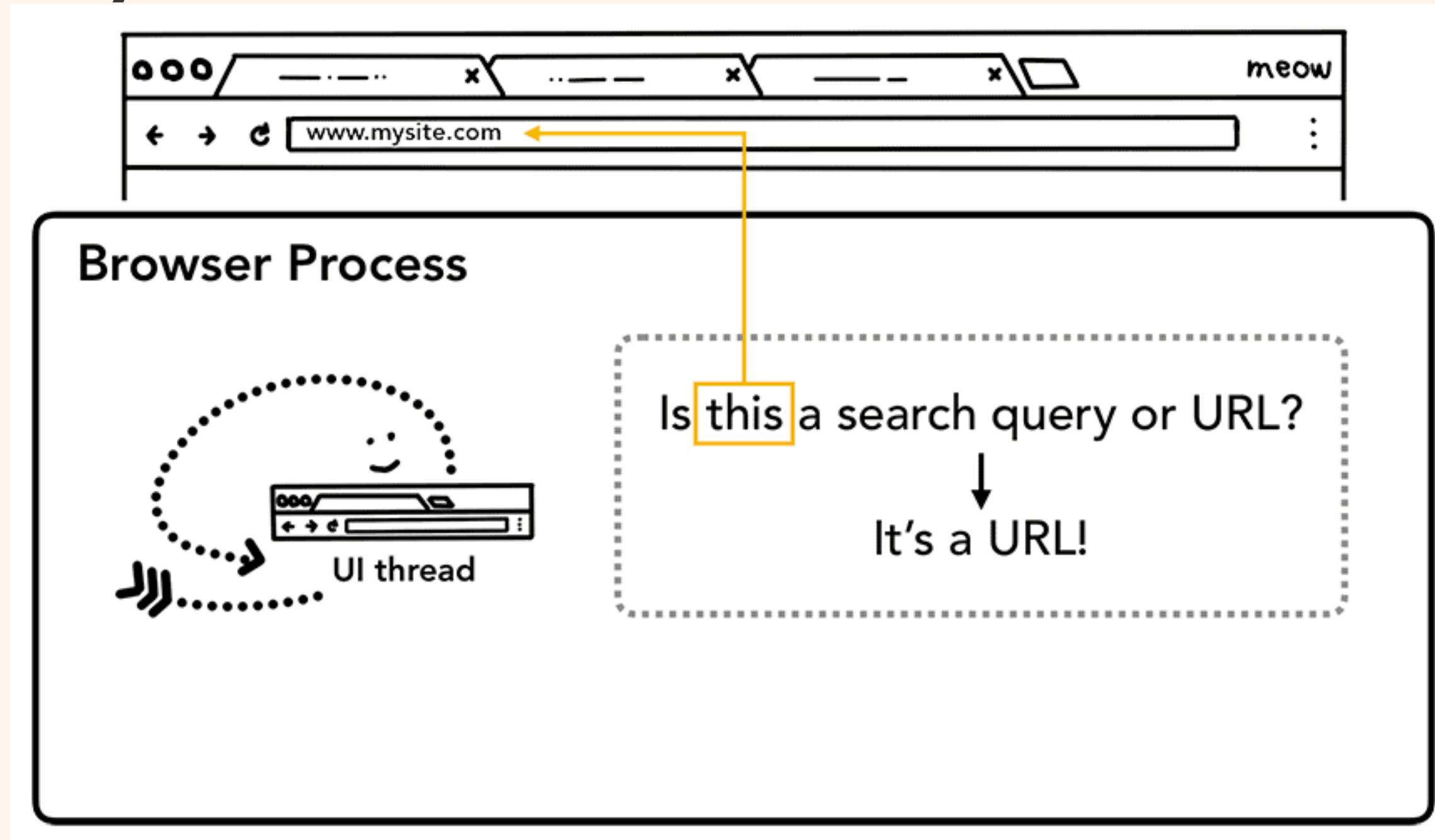


CRITICAL RENDERING PATH

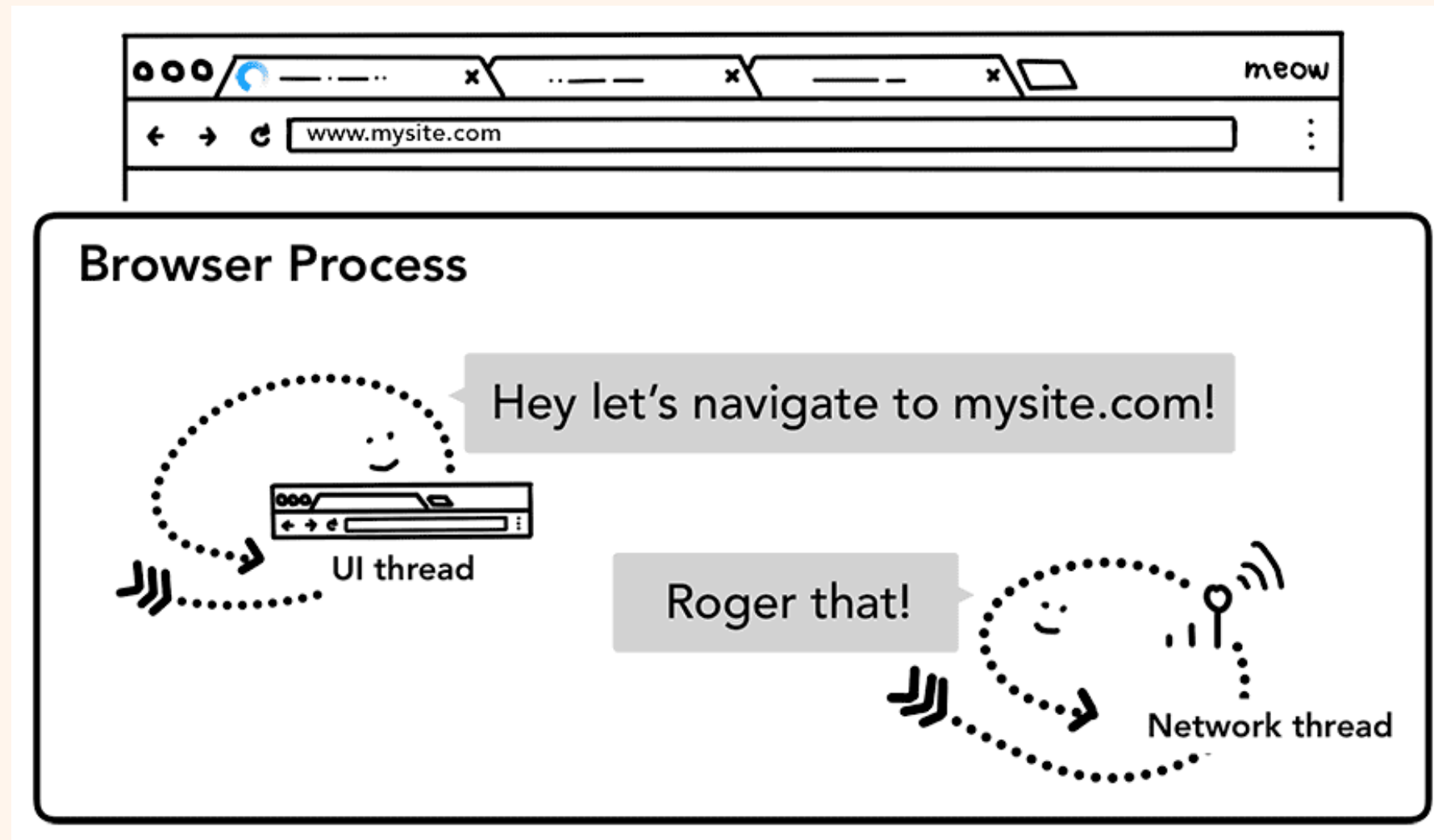
1. Get the initial HTML 
2. Parse the HTML into the DOM
3. Parse CSS into the CSSOM
4. Create render tree
5. Determine where render tree nodes should appear on the page (layout)
6. Paint nodes onto screen (rasterize)

1. GETTING THE INITIAL HTML

In your browser, navigate to a website of your choice



1. GETTING THE INITIAL HTML

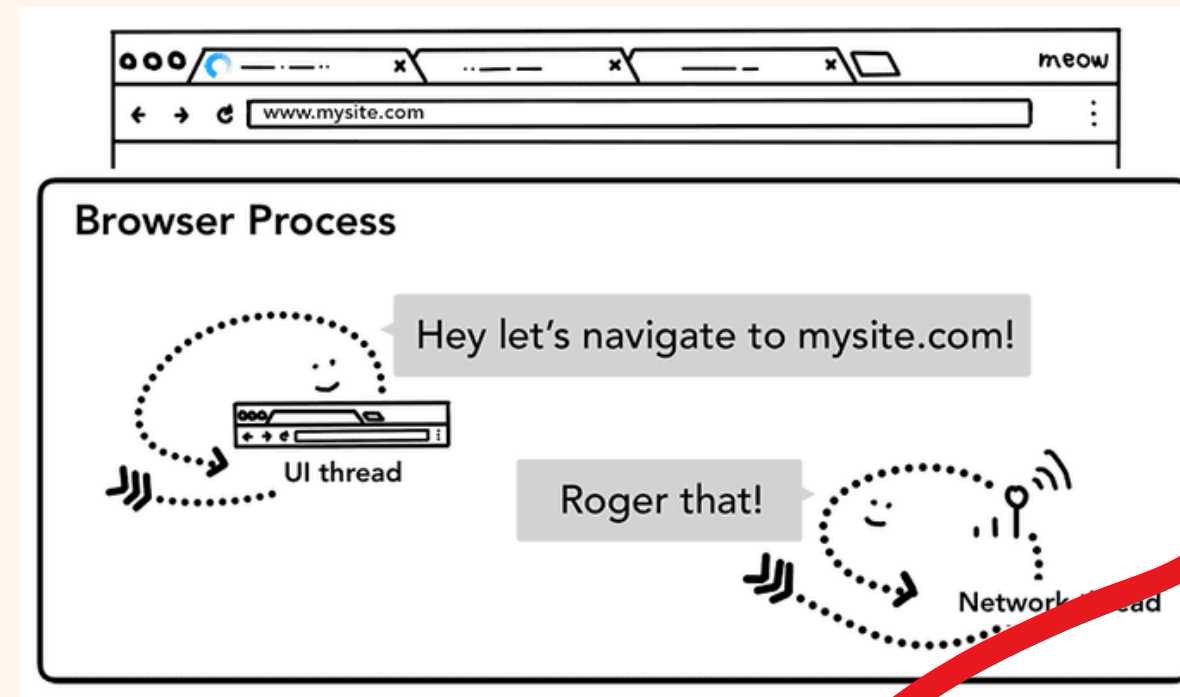


Network
request



Server
(aka Backend or BE)

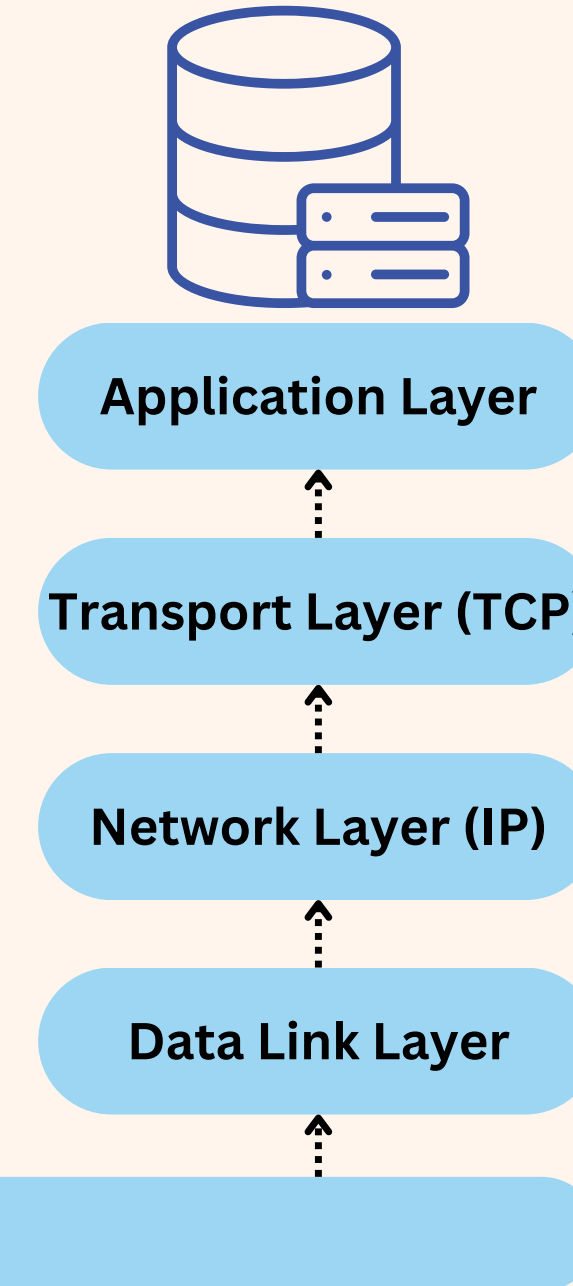
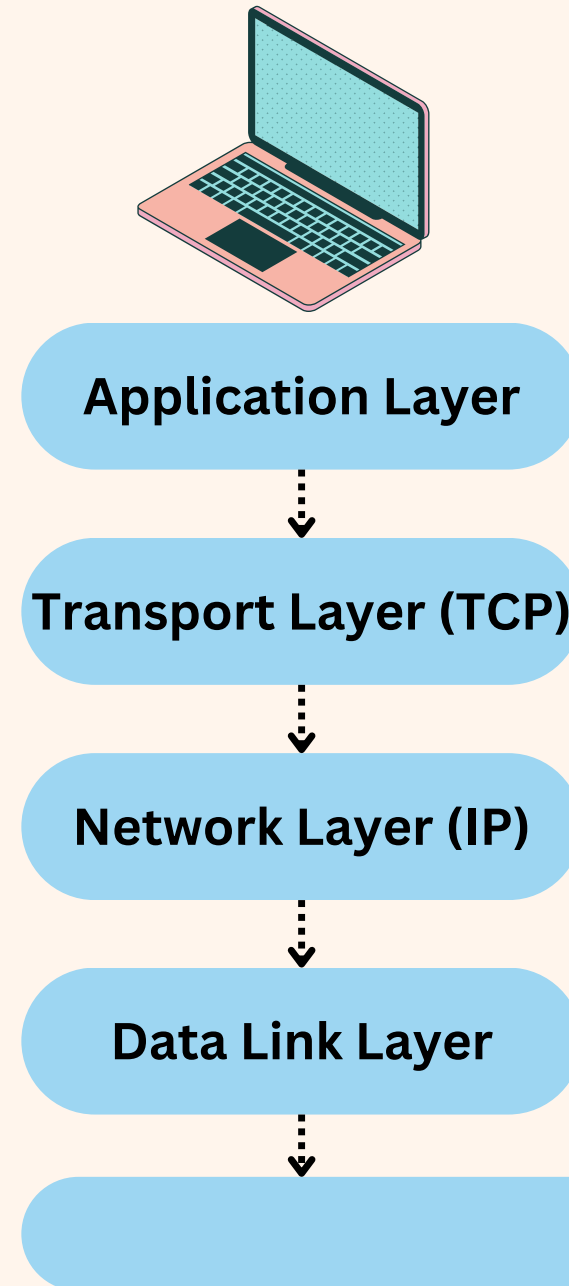
1. GETTING THE INITIAL HTML



Network request

Server
(aka Backend or BE)

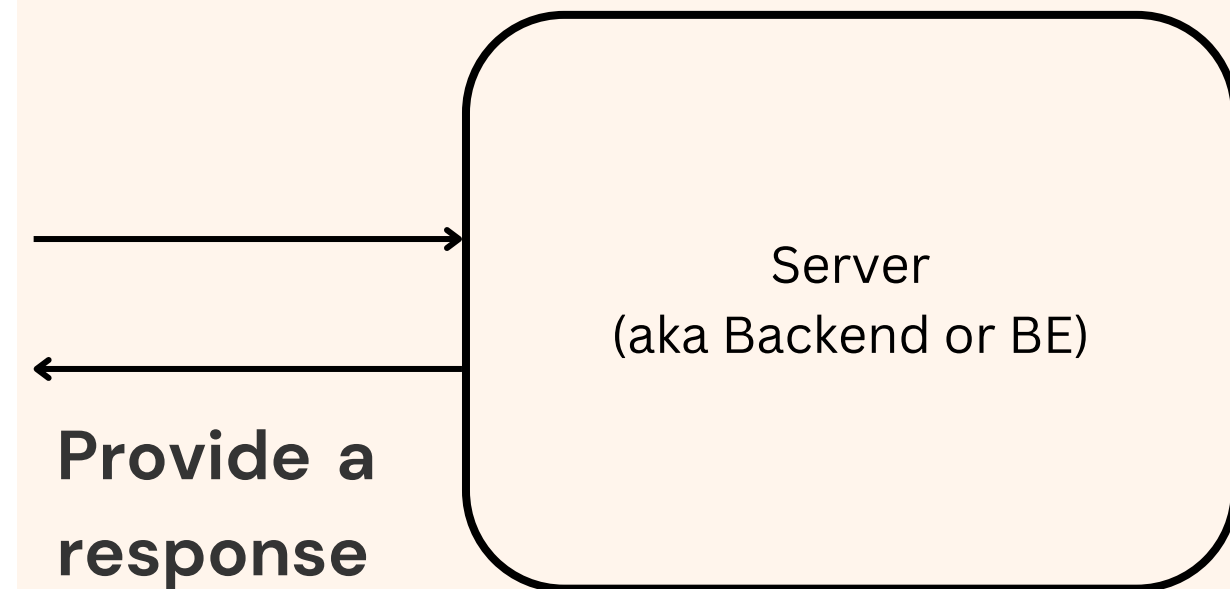
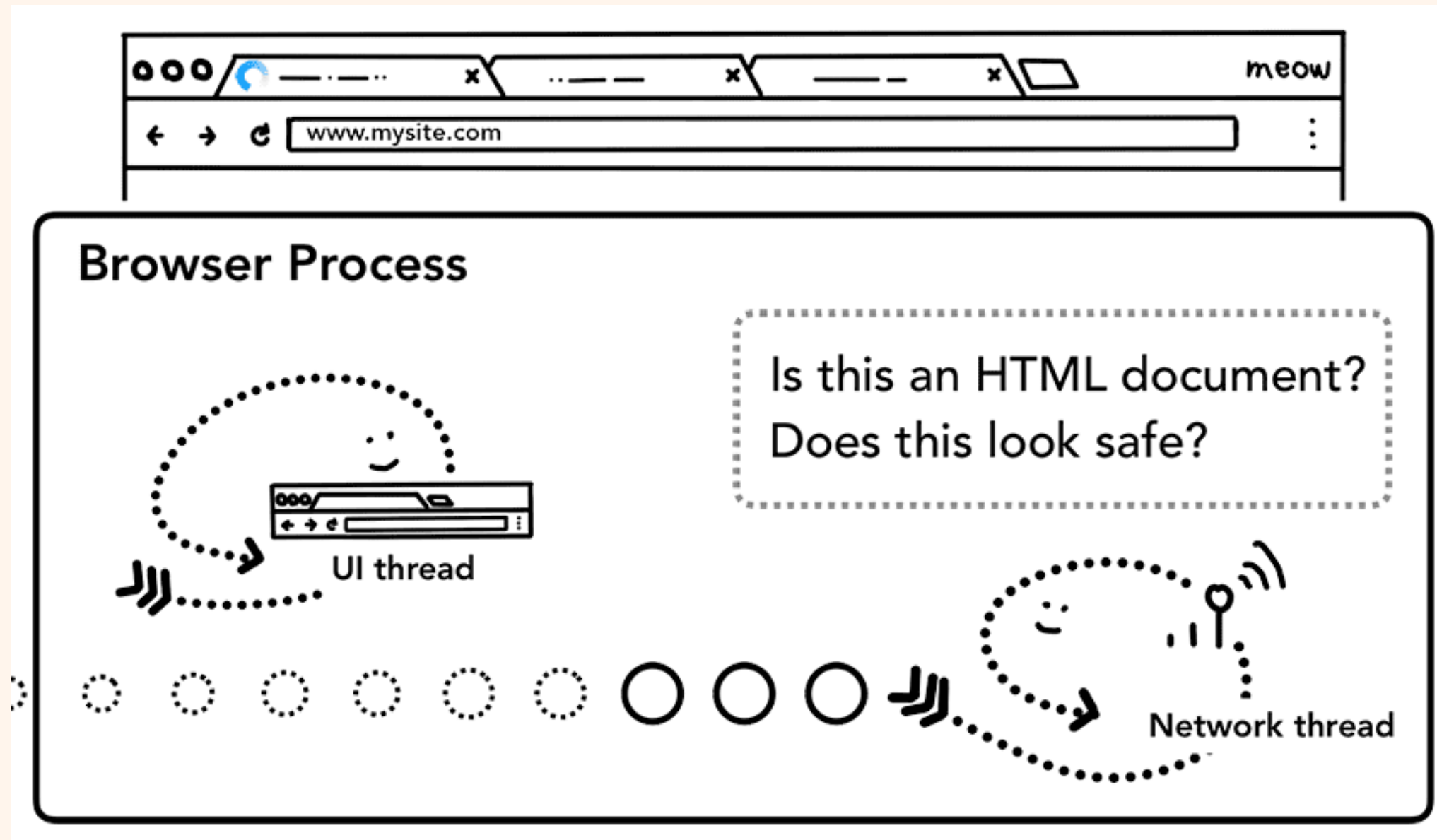
Encapsulation



Decapsulation

Physical Layer (Cables, etc)

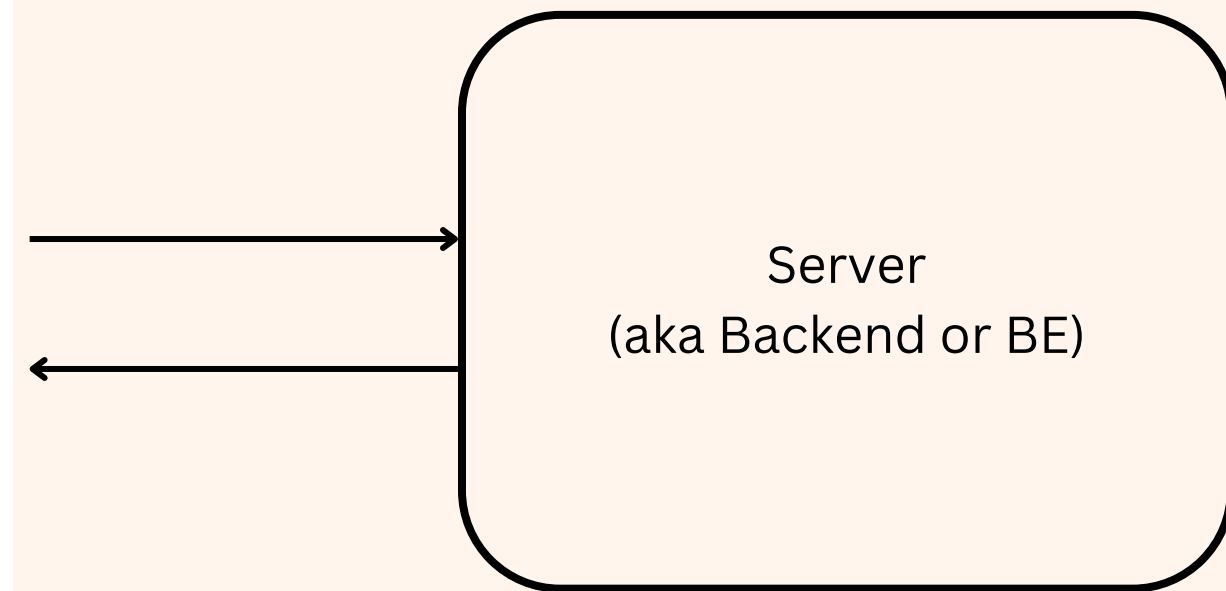
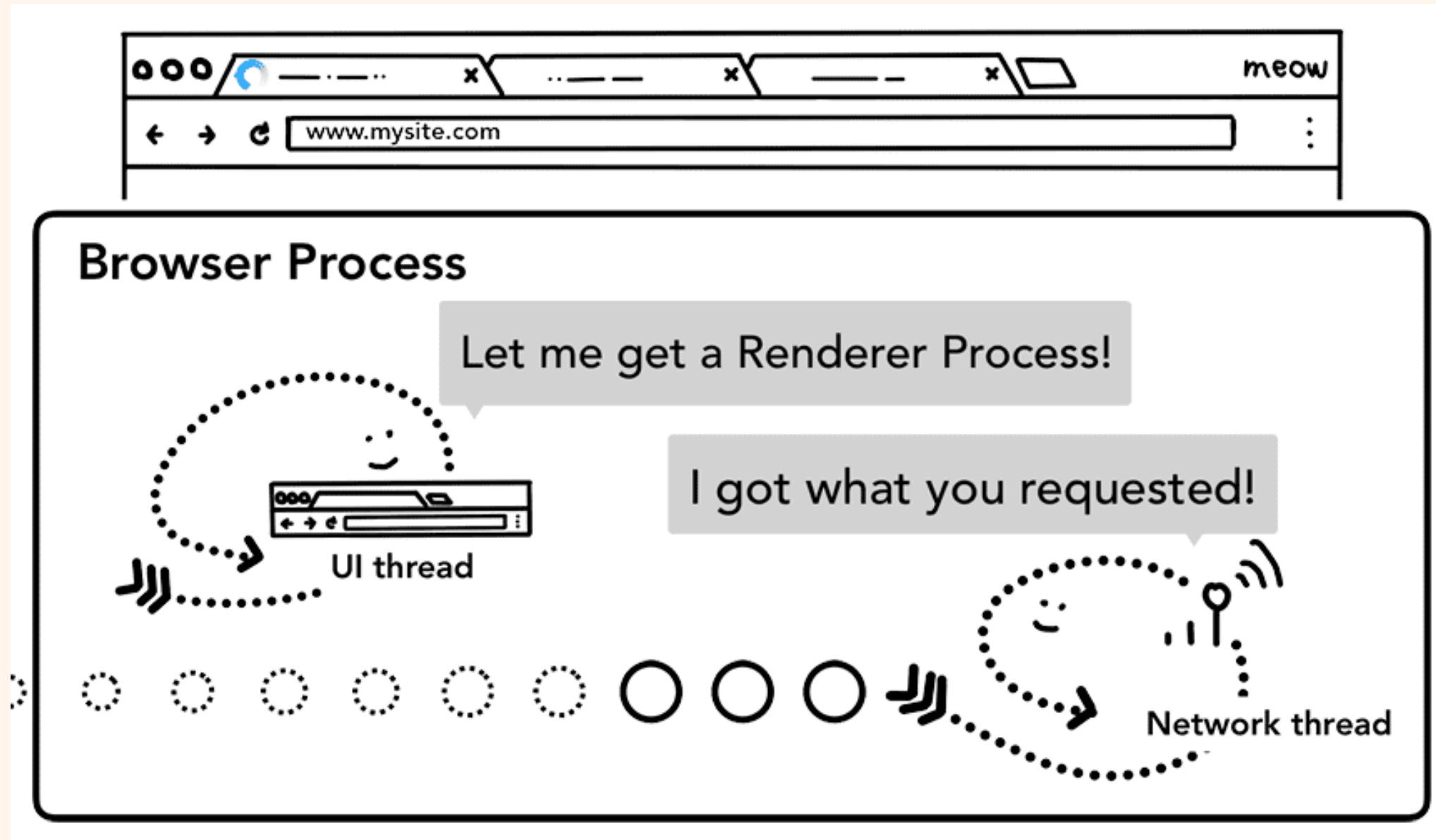
1. GETTING THE INITIAL HTML



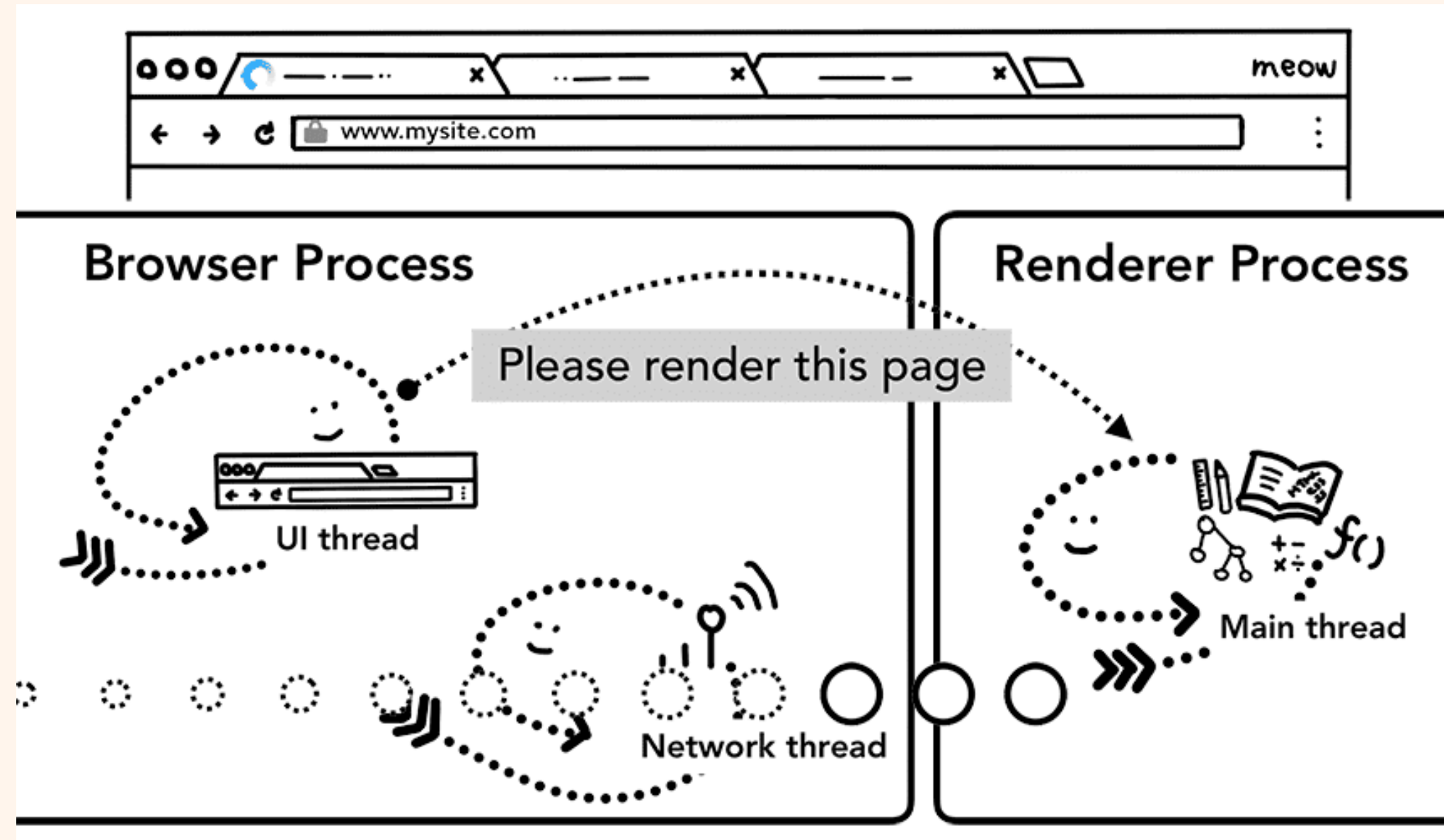
```
header
Content-Length: 68137
Content-Type: text/html

payload
a0 8b 28 3f 01 00 00 00 50
32 68 65 6c 6c 6f 20 77 6f
.....
.....
.....
.....
```

1. GETTING THE INITIAL HTML



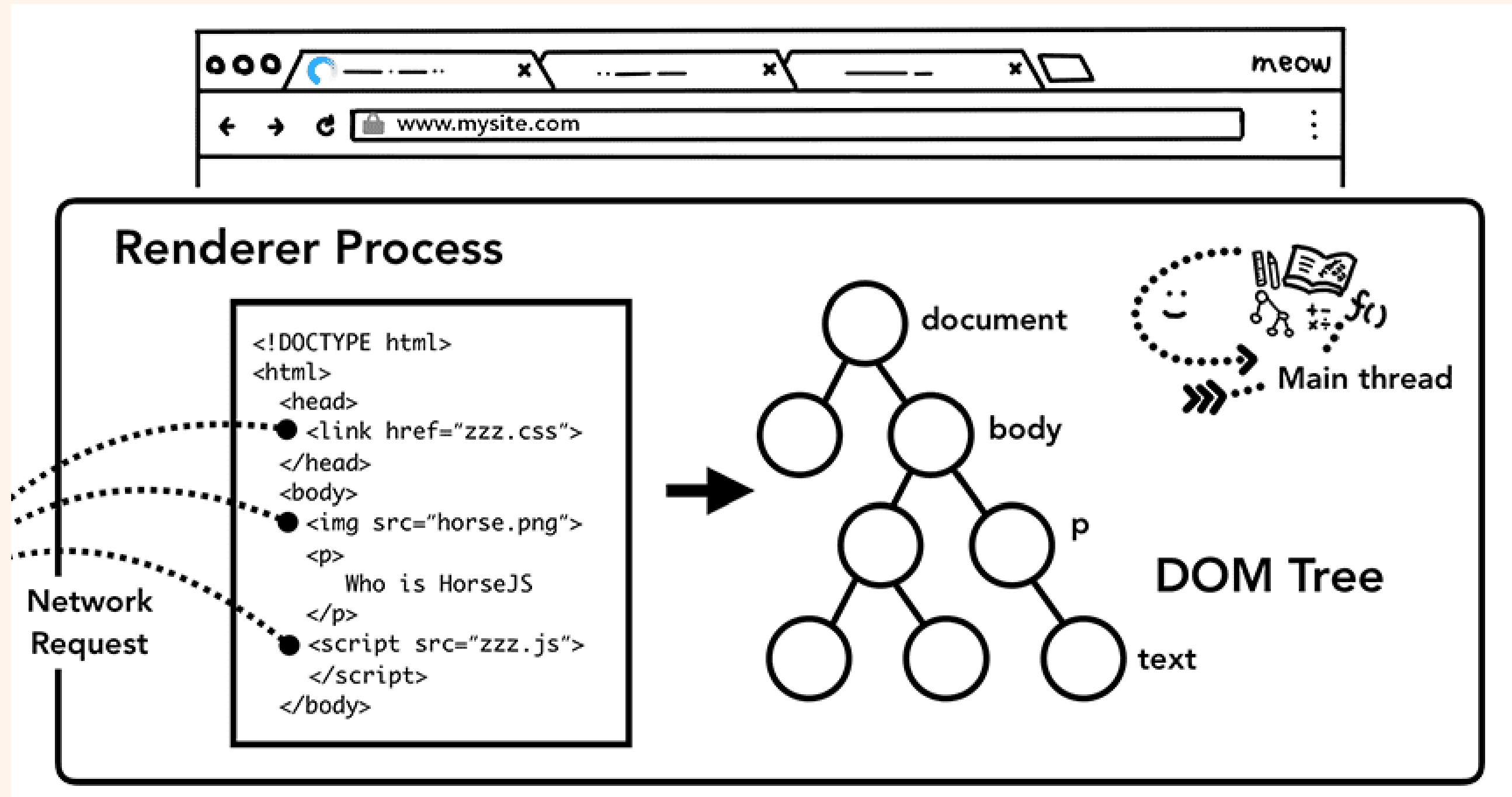
1. GETTING THE INITIAL HTML



CRITICAL RENDERING PATH

1. Get the initial HTML
2. Parse the HTML into the DOM 
3. Parse CSS into the CSSOM
4. Create render tree
5. Determine where render tree nodes should appear on the page (layout)
6. Paint nodes onto screen (rasterize)

2. PARSE THE HTML INTO THE DOM

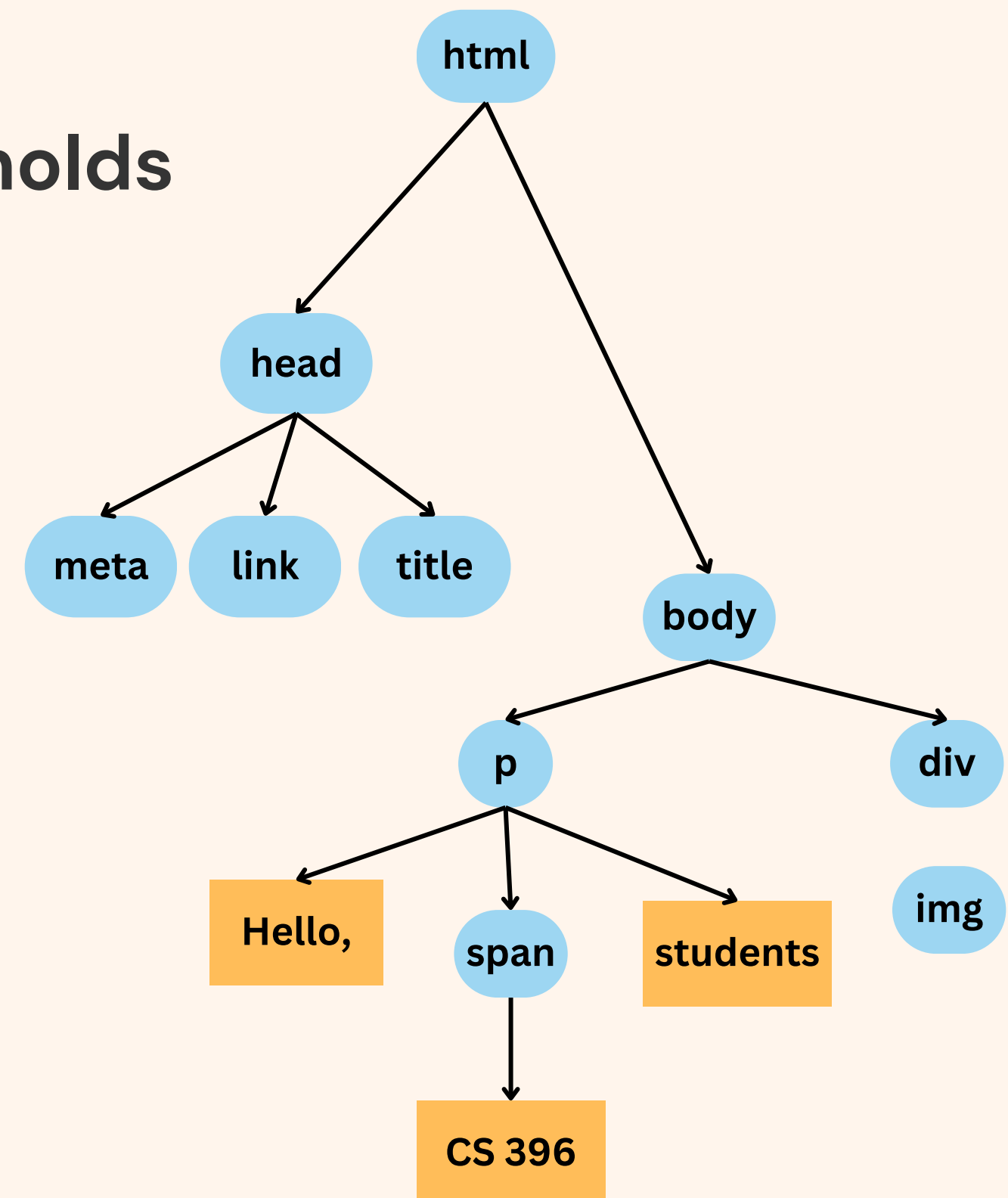


2. PARSE THE HTML INTO THE DOM

DOM = Document Object Model

- Browser's tree data structure that holds what goes into a page

```
<!DOCTYPE html>
<html>
  <head>
    <meta name="viewport" content="width=device-width,initial-scale=1" />
    <link href="style.css" rel="stylesheet" />
    <title>DOM + CSSOM Trees</title>
  </head>
  <body>
    <p>Hello <span>CS 396</span> students</p>
    <div></div>
  </body>
</html>
```



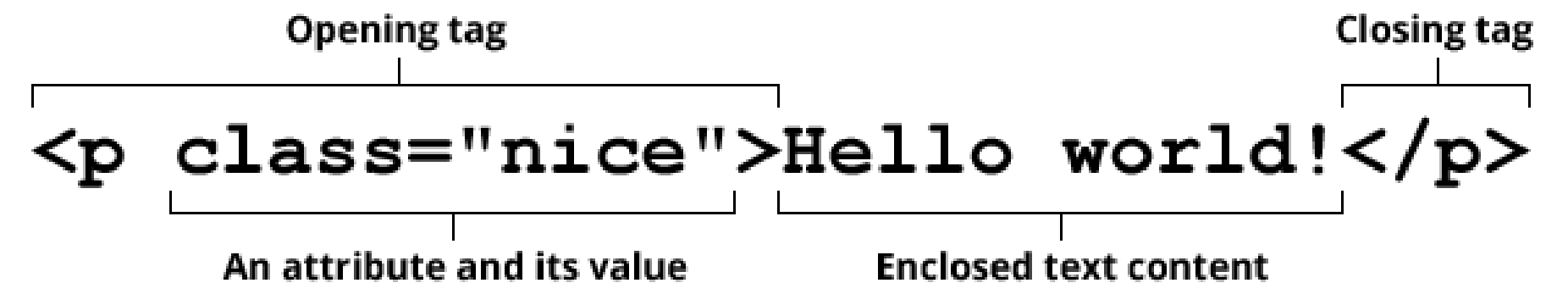
2. PARSE THE HTML INTO THE DOM

Your turn:

1. Fork stackblitz.com/edit/cs303html
2. Write your own html

```
<!DOCTYPE html>
<html>
  <head>
    <meta name="viewport" content="width=device-width,initial-scale=1" />
    <link href="style.css" rel="stylesheet" />
    <title>DOM + CSSOM Trees</title>
  </head>
  <body>
    <p>Hello <span>CS 396</span> students</p>
    <div></div>
  </body>
</html>
```

Anatomy of an HTML element

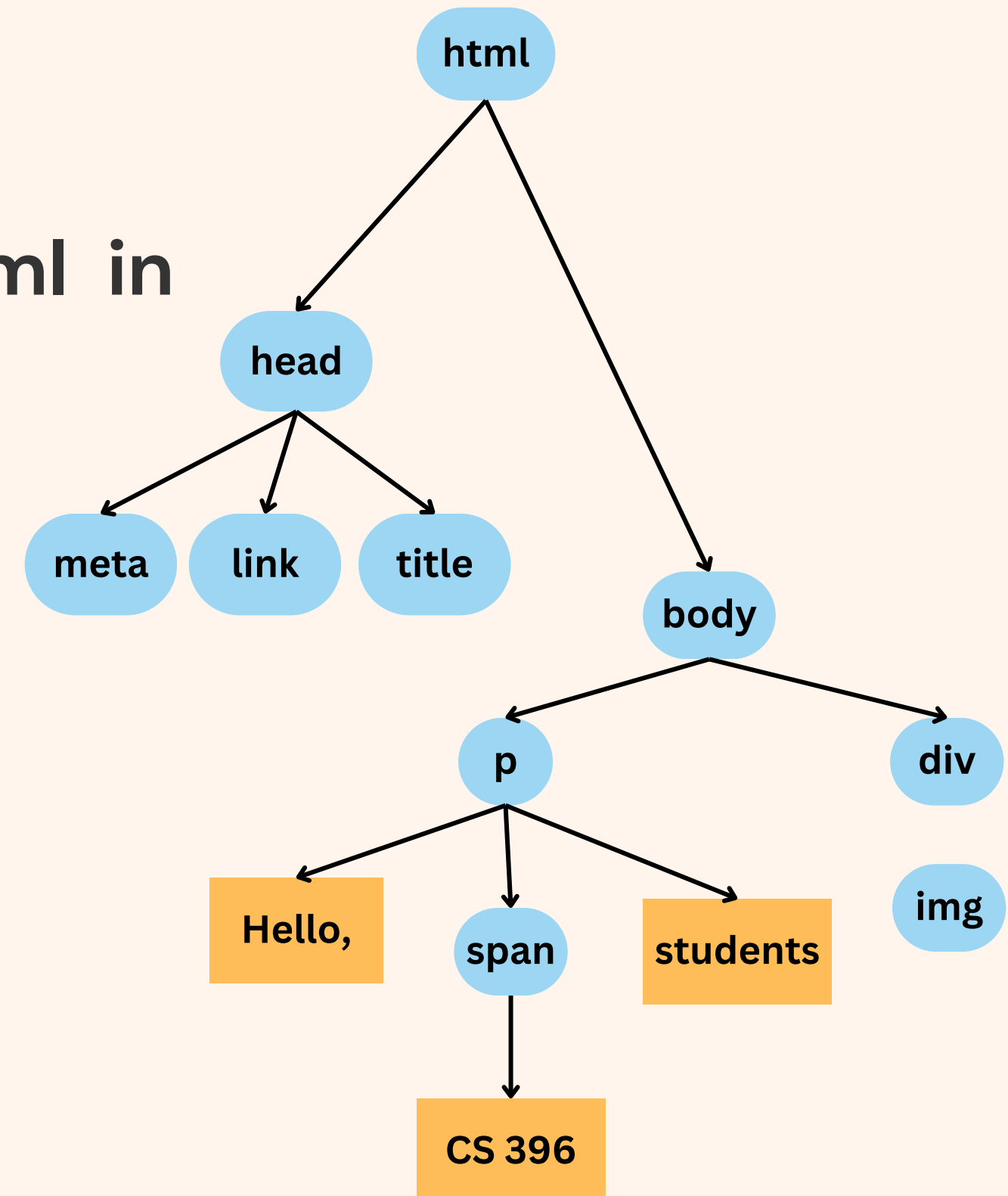


2. PARSE THE HTML INTO THE DOM

Collaborate:

1. Trade your stackblitz with a partner
2. Sketch the DOM for their index.html in FigJam

```
<!DOCTYPE html>
<html>
  <head>
    <meta name="viewport" content="width=device-width,initial-scale=1" />
    <link href="style.css" rel="stylesheet" />
    <title>DOM + CSSOM Trees</title>
  </head>
  <body>
    <p>Hello <span>CS 396</span> students</p>
    <div></div>
  </body>
</html>
```

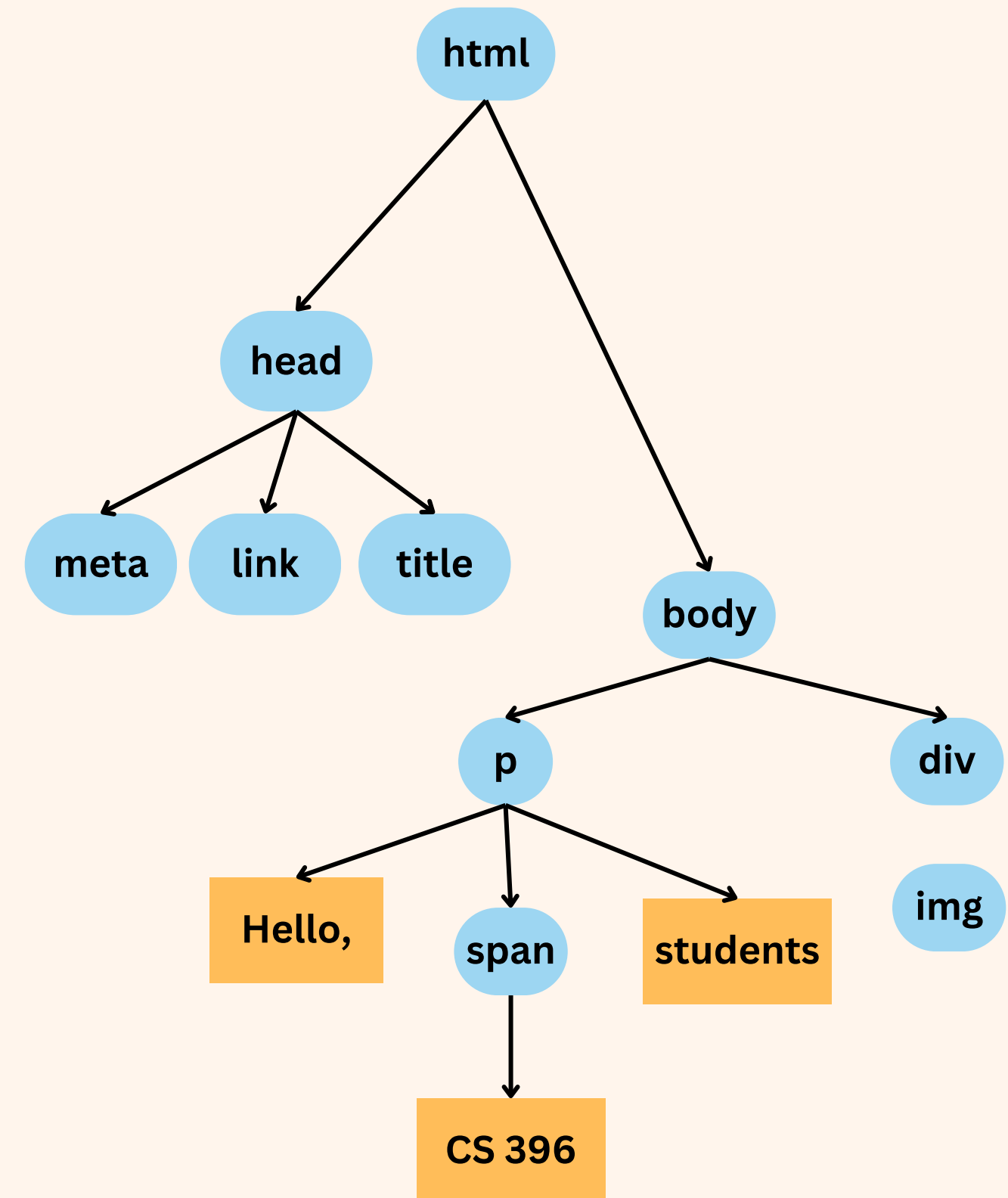


2. PARSE THE HTML INTO THE DOM

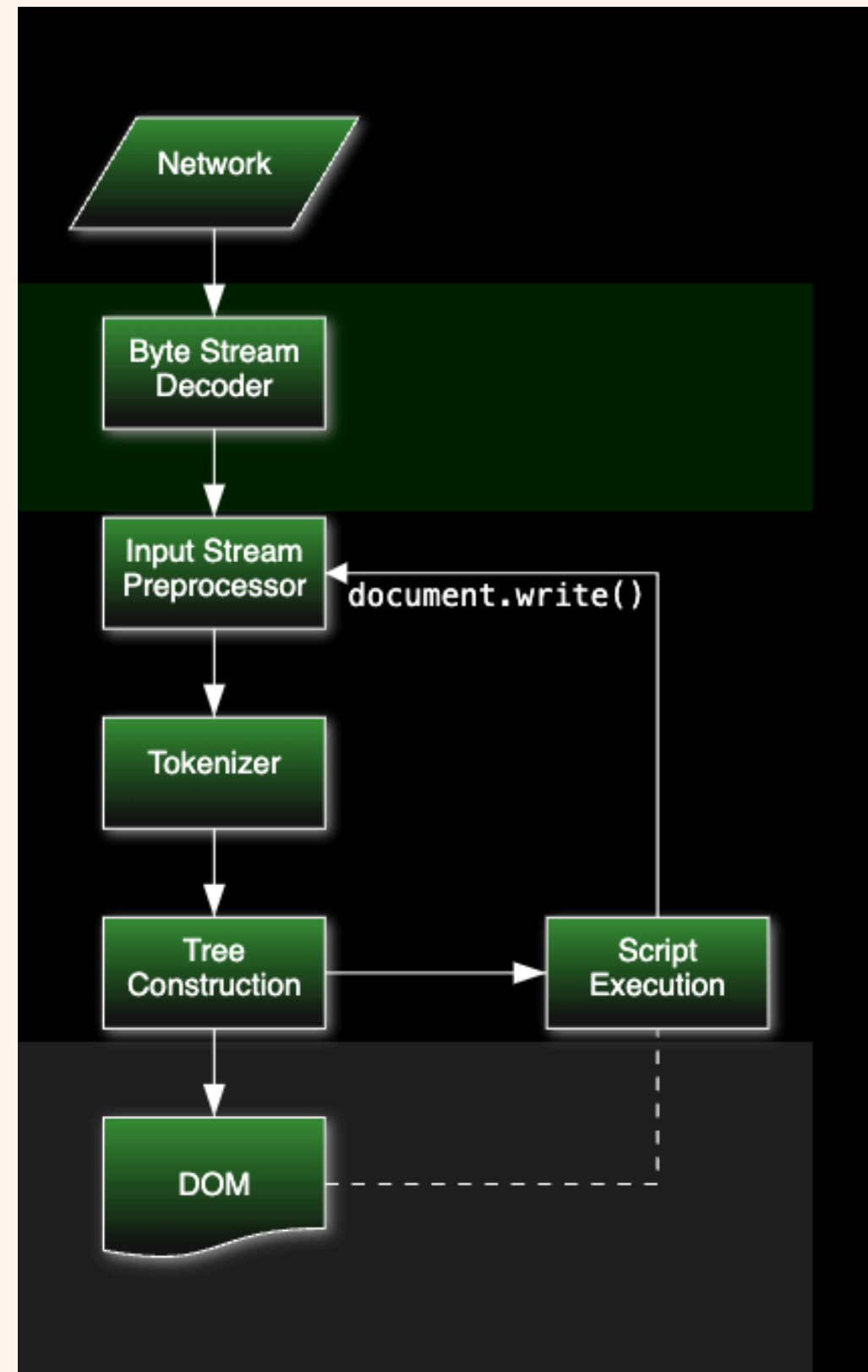
Documentation (in your learning doc):

1. Add your stackblitz link
2. Add a screenshot of the DOM sketch

```
<!DOCTYPE html>
<html>
  <head>
    <meta name="viewport" content="width=device-width,initial-scale=1" />
    <link href="style.css" rel="stylesheet" />
    <title>DOM + CSSOM Trees</title>
  </head>
  <body>
    <p>Hello <span>CS 396</span> students</p>
    <div></div>
  </body>
</html>
```



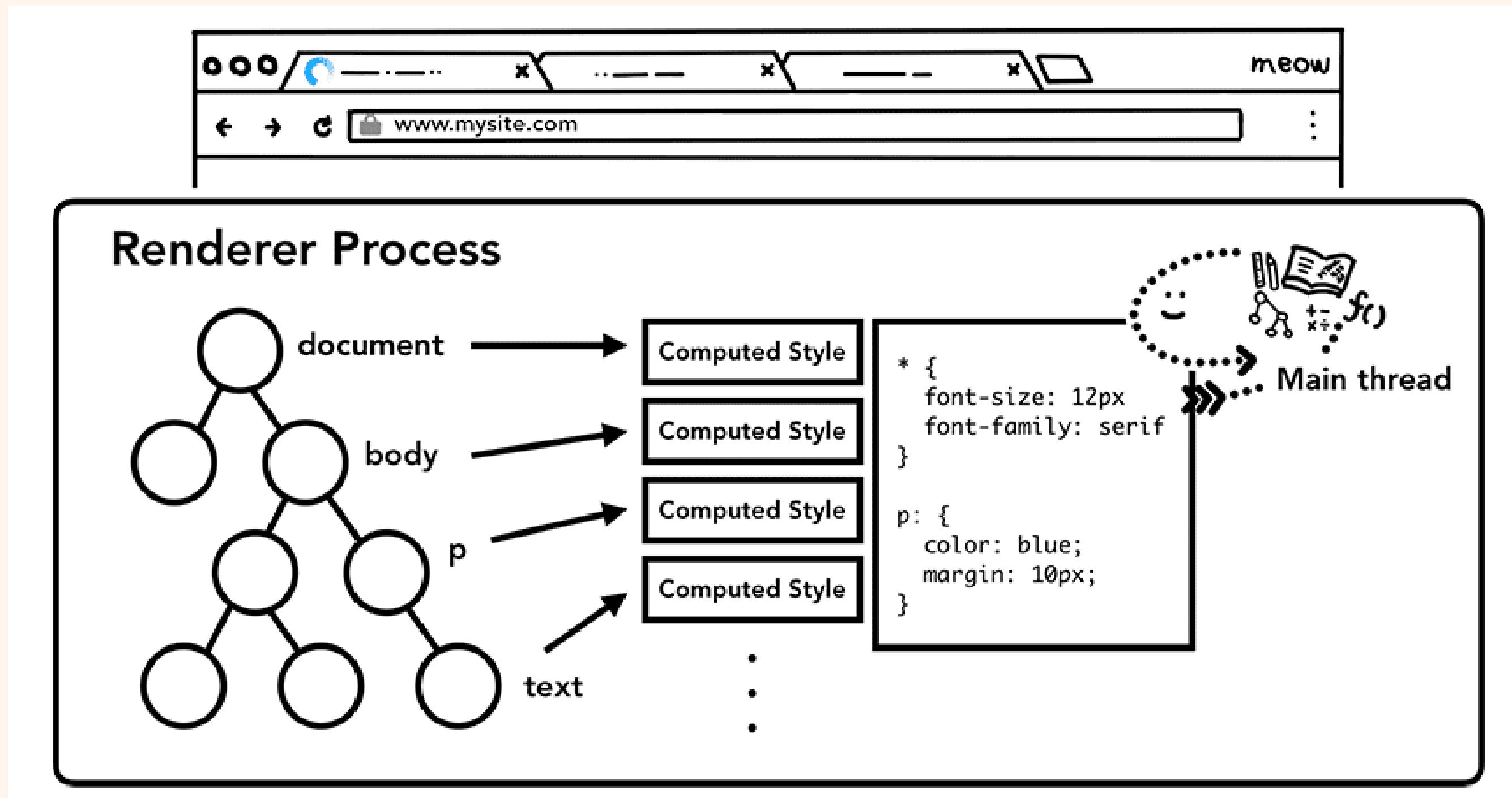
2. PARSE THE HTML INTO THE DOM



CRITICAL RENDERING PATH

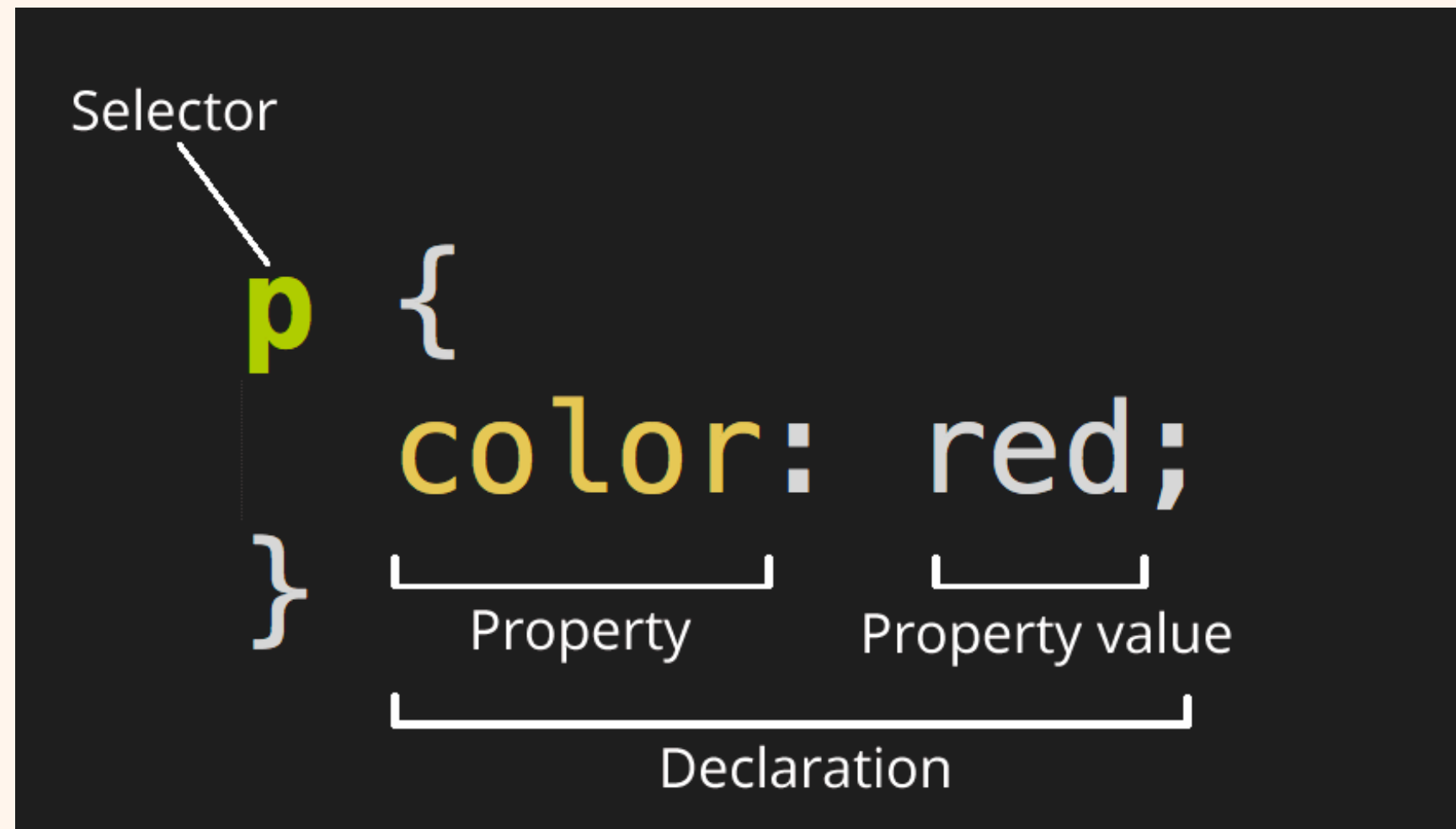
1. Get the initial HTML
2. Parse the HTML into the DOM
3. Parse CSS into the CSSOM 
4. Create render tree
5. Determine where render tree nodes should appear on the page (layout)
6. Paint nodes onto screen (rasterize)

3. PARSE CSS INTO THE CSSOM



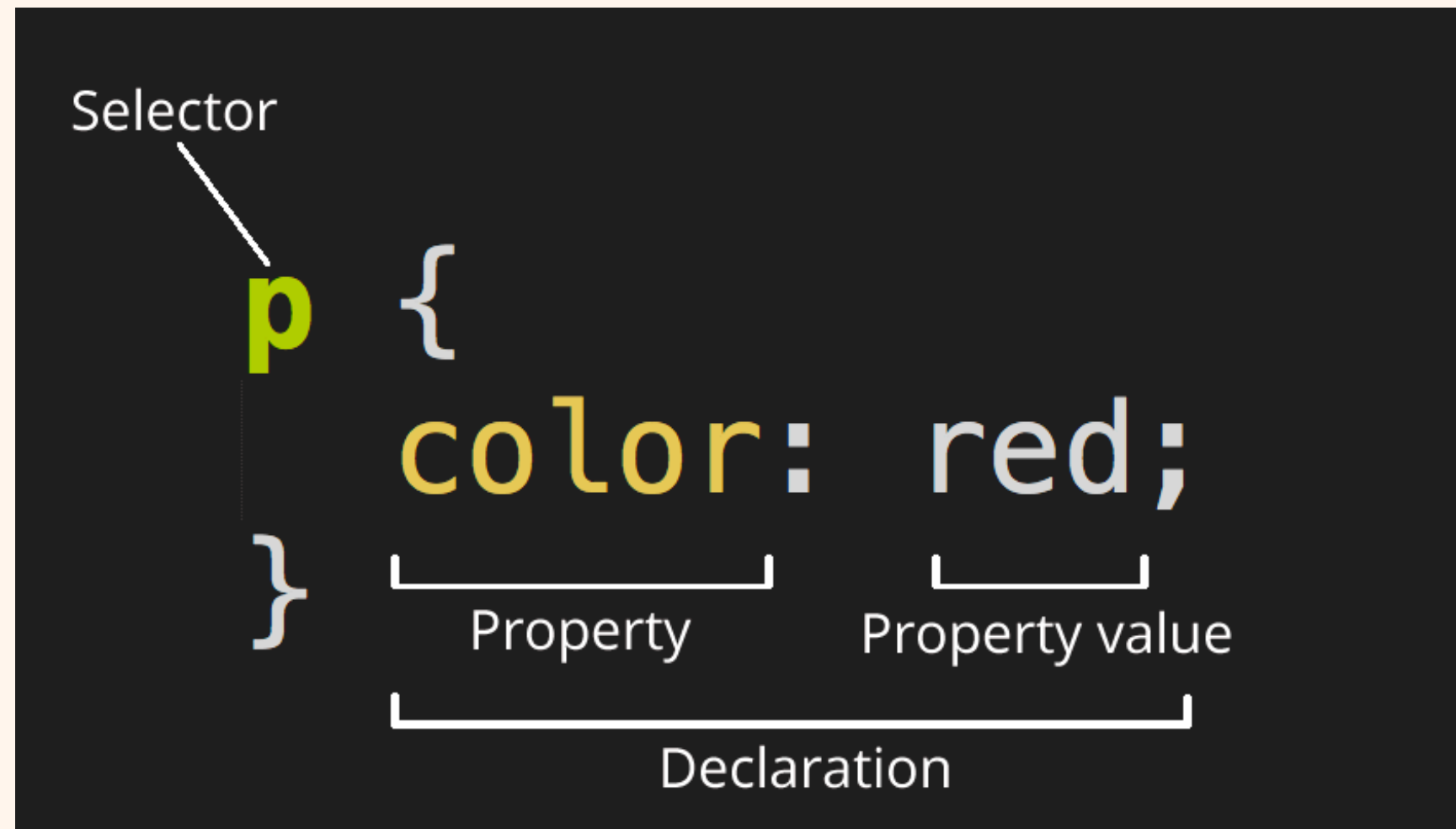
3. PARSE CSS INTO THE CSSOM

```
body {  
  font-size: 16px;  
}  
p {  
  font-weight: bold;  
}  
span {  
  color: red;  
}  
p span {  
  display: none;  
}  
img {  
  float: right;  
}
```



3. PARSE CSS INTO THE CSSOM

```
body {  
  font-size: 16px;  
}  
p {  
  font-weight: bold;  
}  
span {  
  color: red;  
}  
p span {  
  display: none;  
}  
img {  
  float: right;  
}
```

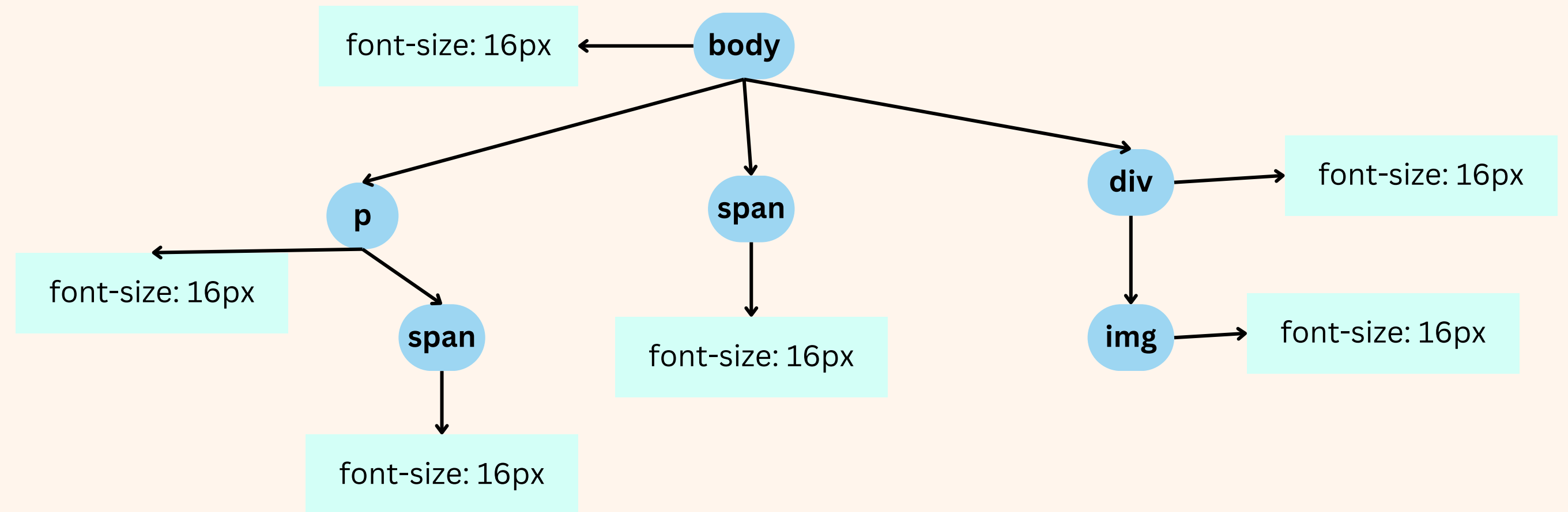


Your turn:

- Choose an element in your index.html to add styling

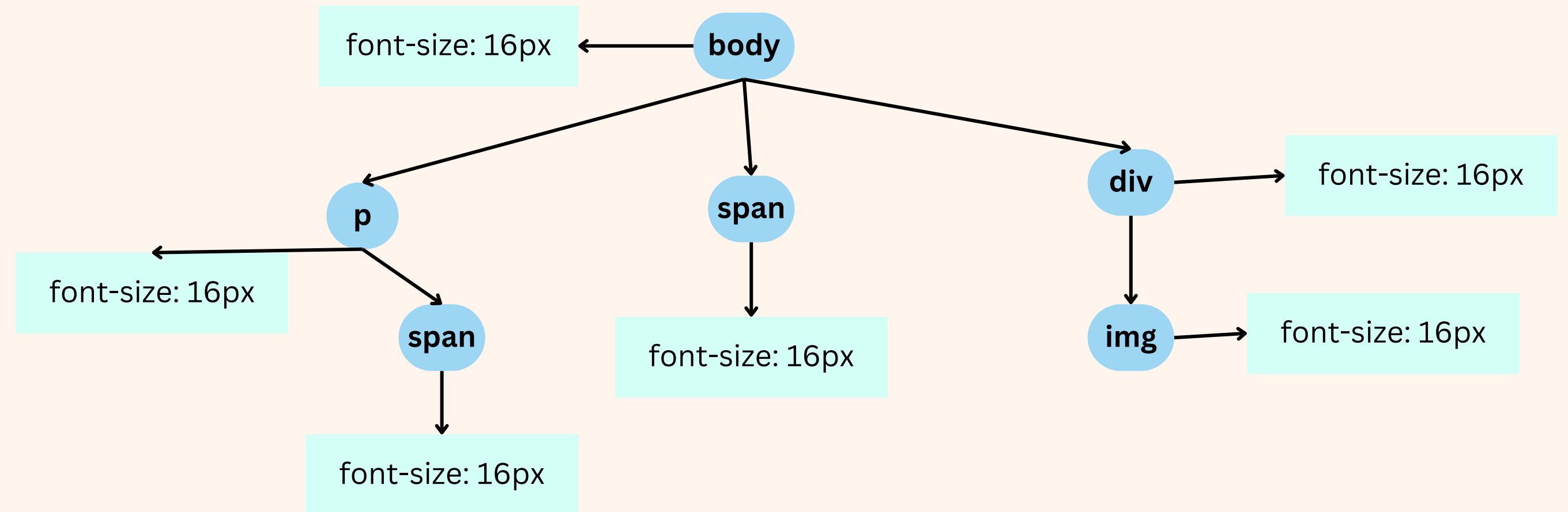
3. PARSE CSS INTO THE CSSOM

```
body {  
  font-size: 16px;  
}  
p {  
  font-weight: bold;  
}  
span {  
  color: red;  
}  
p span {  
  display: none;  
}  
img {  
  float: right;  
}
```



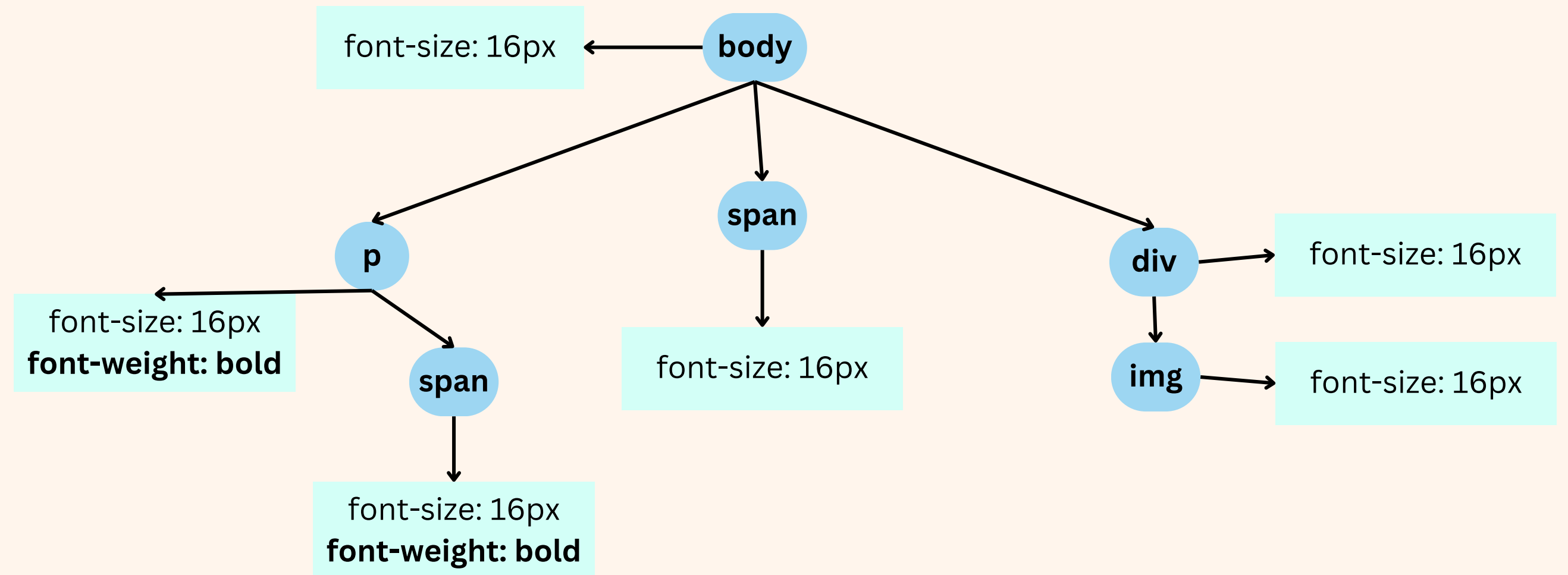
3. PARSE CSS INTO THE CSSOM

```
body {  
  font-size: 16px;  
}  
p {  
  font-weight: bold;  
}  
span {  
  color: red;  
}  
p span {  
  display: none;  
}  
img {  
  float: right;  
}
```



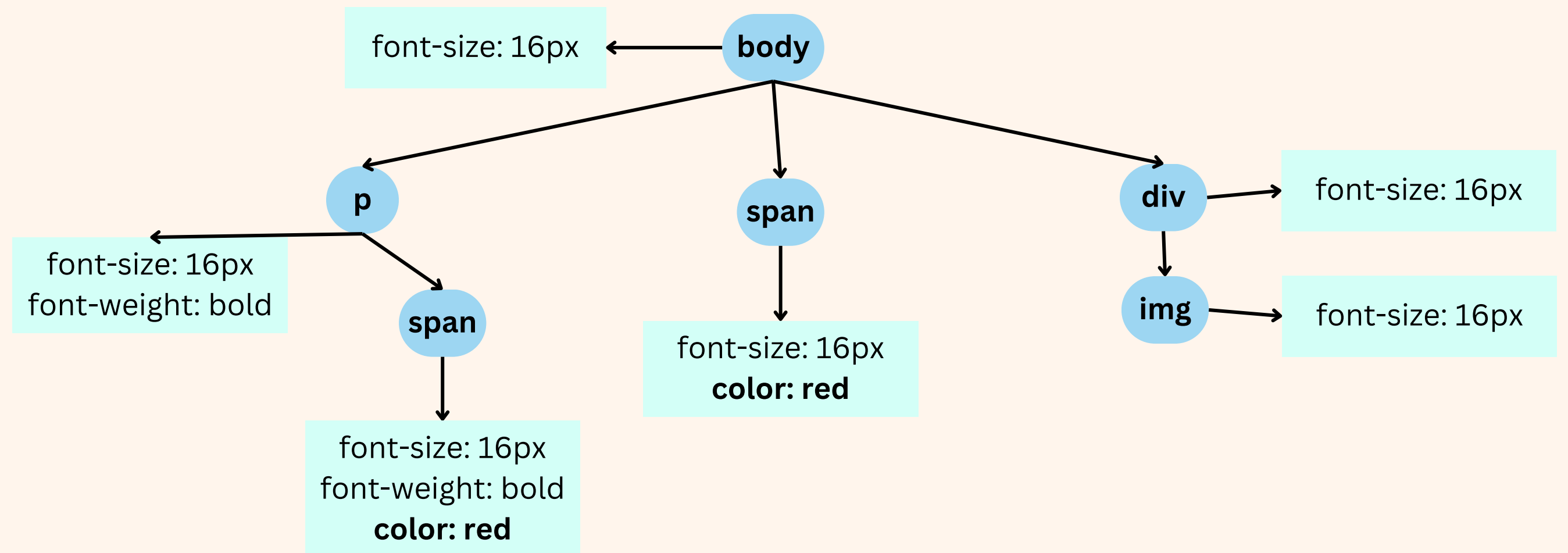
3. PARSE CSS INTO THE CSSOM

```
body {  
  font-size: 16px;  
}  
p {  
  font-weight: bold;  
}  
span {  
  color: red;  
}  
p span {  
  display: none;  
}  
img {  
  float: right;  
}
```



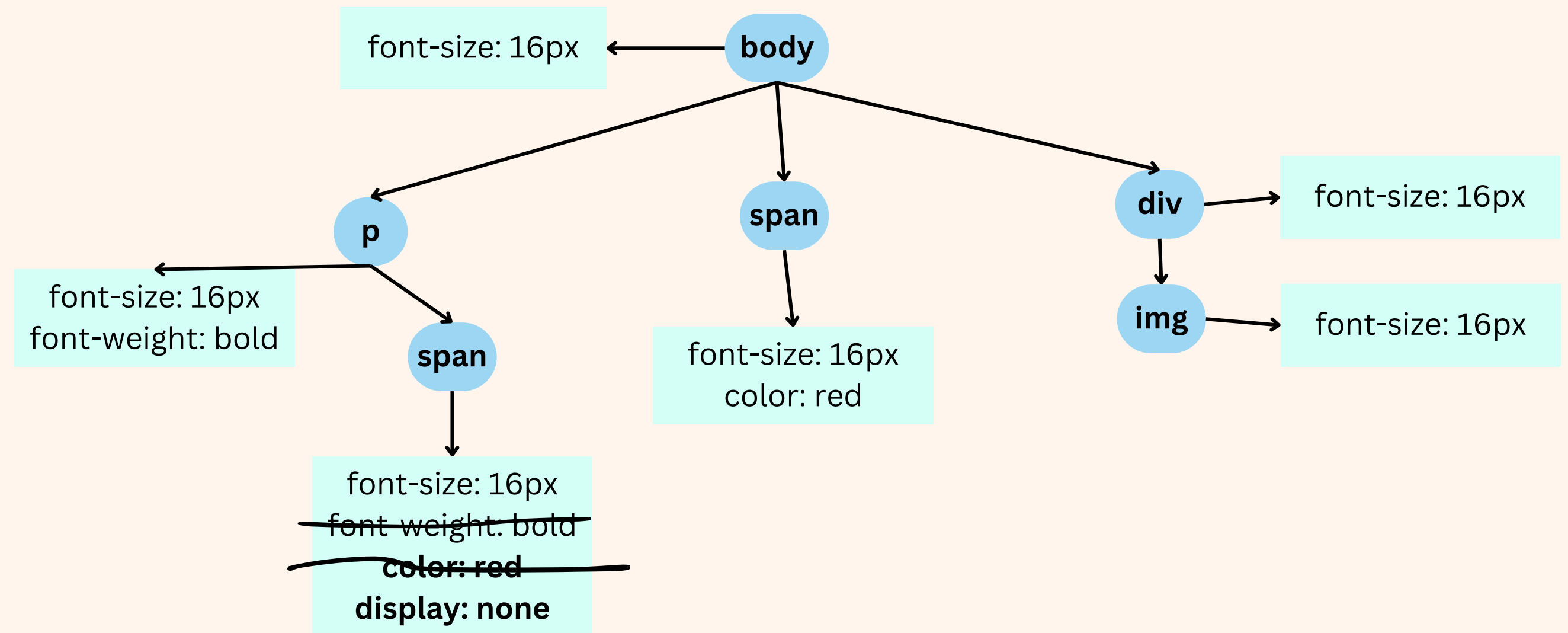
3. PARSE CSS INTO THE CSSOM

```
body {  
  font-size: 16px;  
}  
p {  
  font-weight: bold;  
}  
span {  
  color: red;  
}  
p span {  
  display: none;  
}  
img {  
  float: right;  
}
```



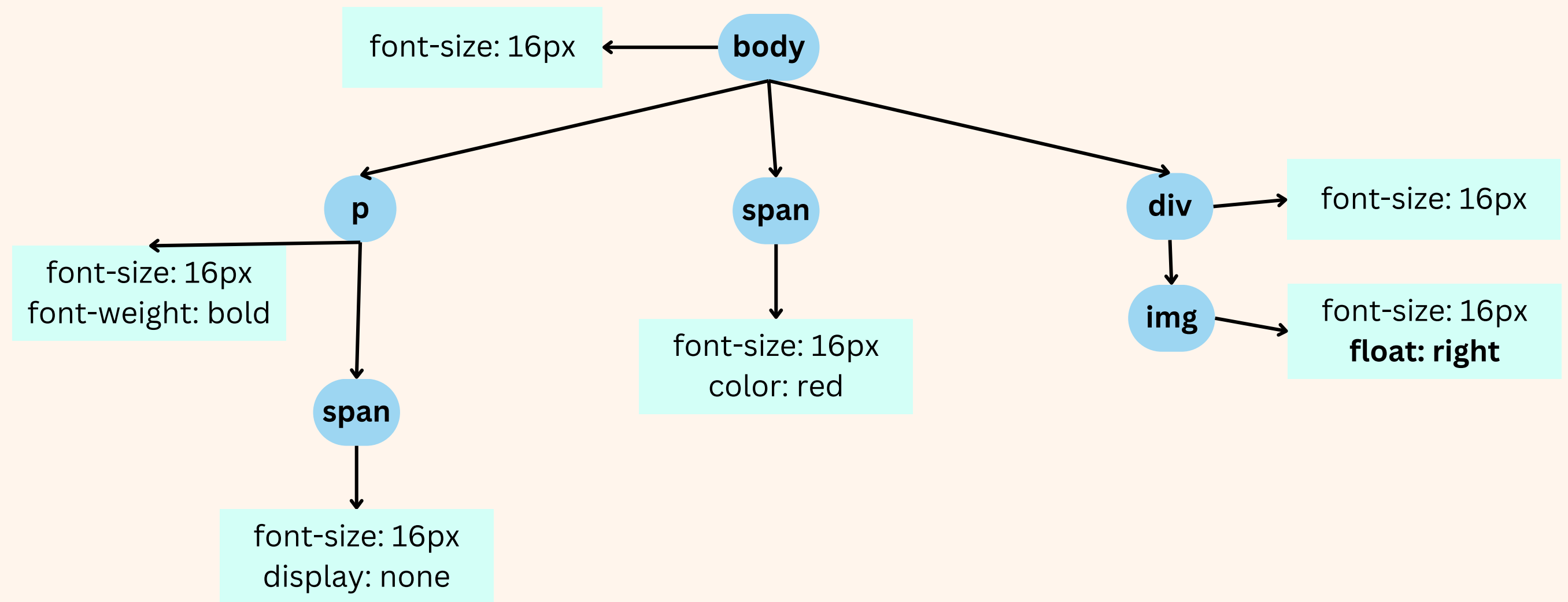
3. PARSE CSS INTO THE CSSOM

```
body {  
  font-size: 16px;  
}  
p {  
  font-weight: bold;  
}  
span {  
  color: red;  
}  
p span {  
  display: none;  
}  
img {  
  float: right;  
}
```



3. PARSE CSS INTO THE CSSOM

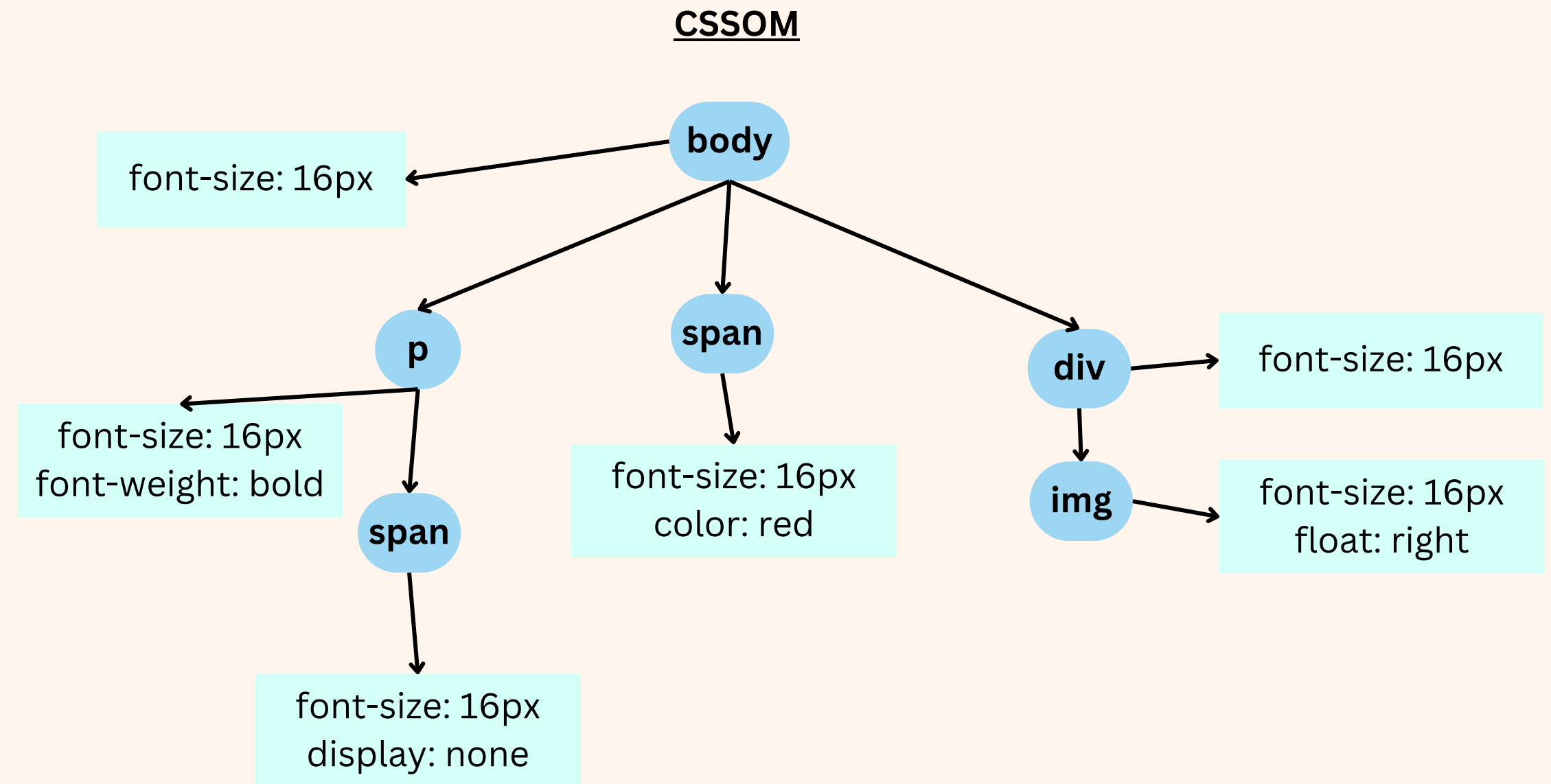
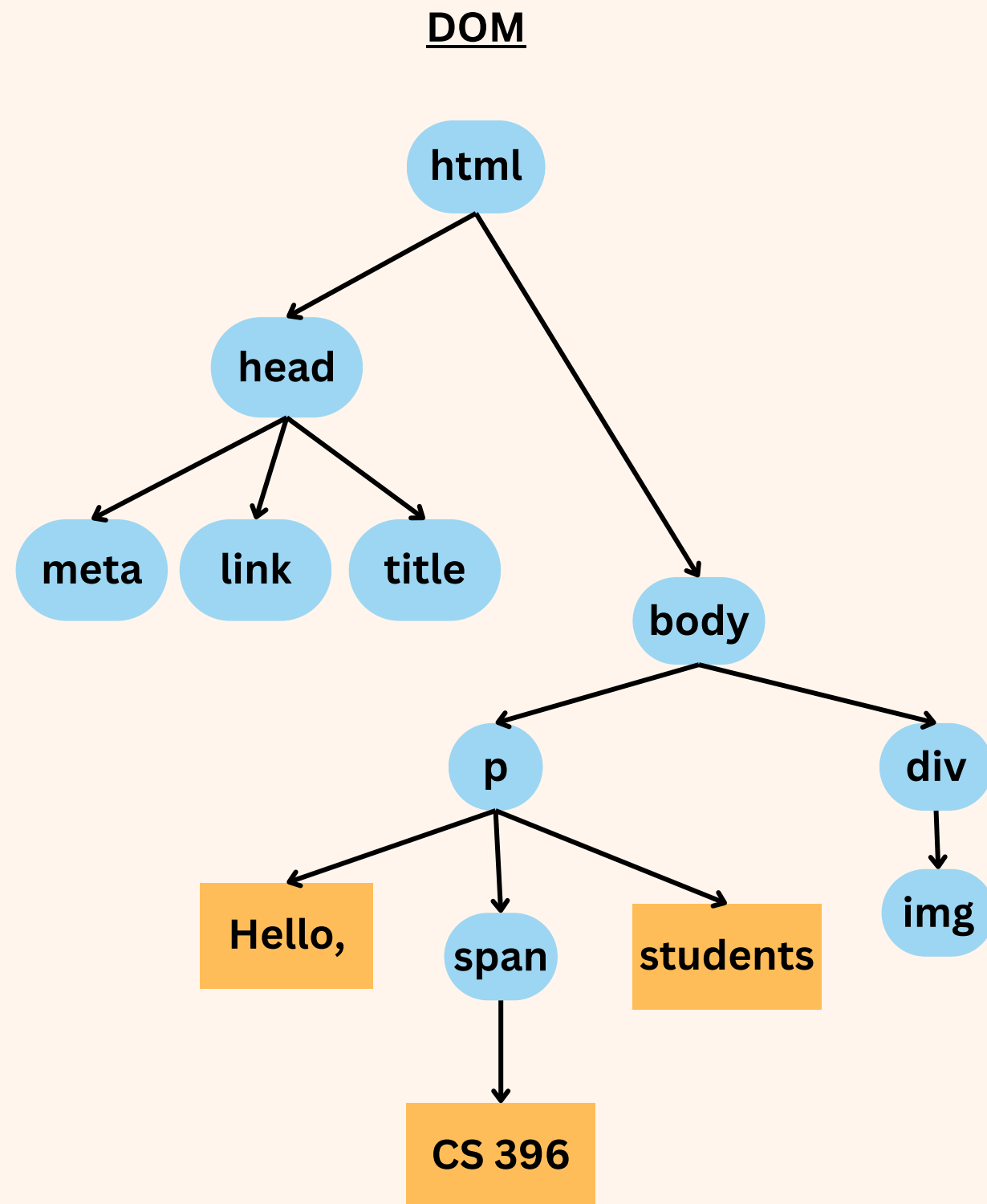
```
body {  
  font-size: 16px;  
}  
p {  
  font-weight: bold;  
}  
span {  
  color: red;  
}  
p span {  
  display: none;  
}  
img {  
  float: right;  
}
```



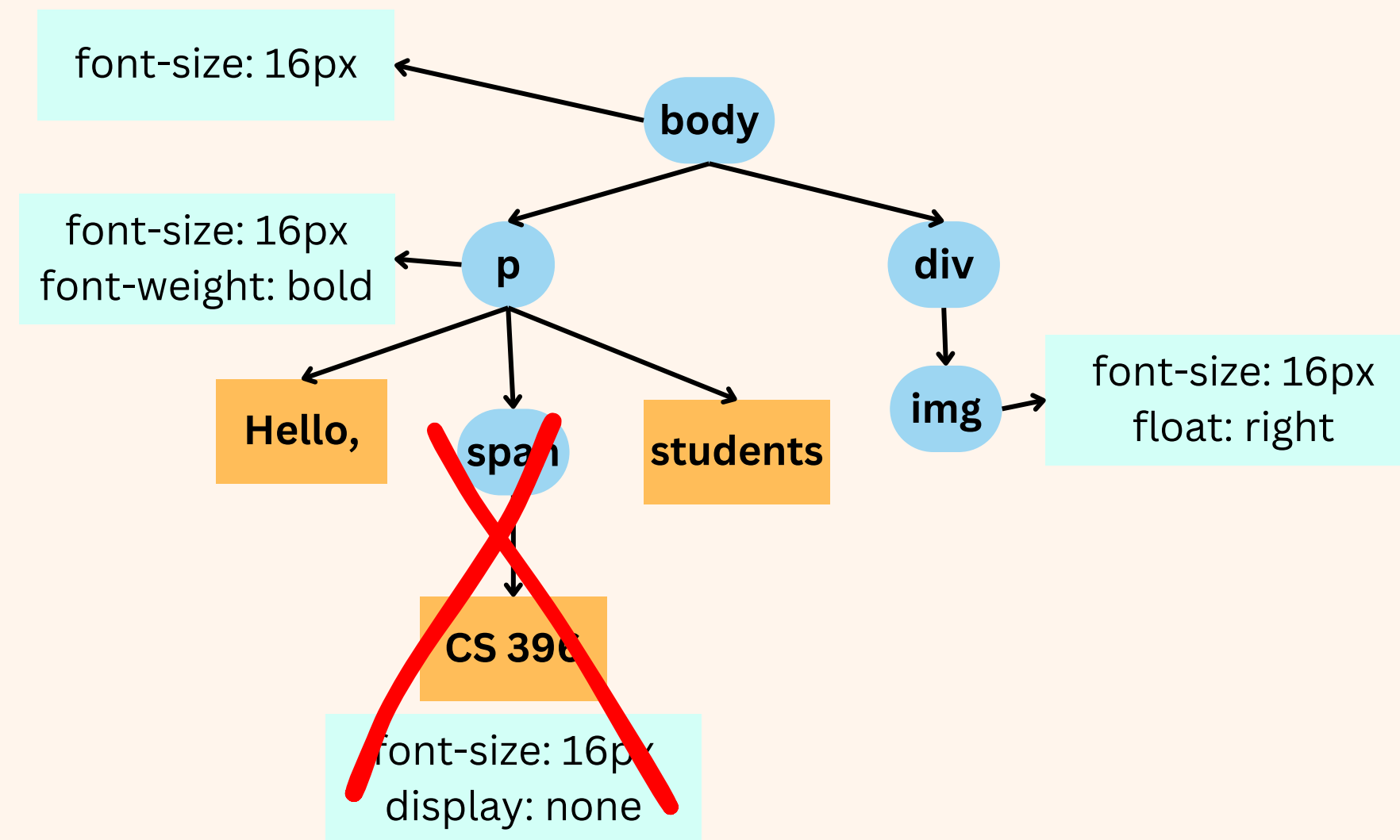
CRITICAL RENDERING PATH

1. Get the initial HTML
2. Parse the HTML into the DOM
3. Parse CSS into the CSSOM
4. Create render tree 
5. Determine where render tree nodes should appear on the page (layout)
6. Paint nodes onto screen (rasterize)

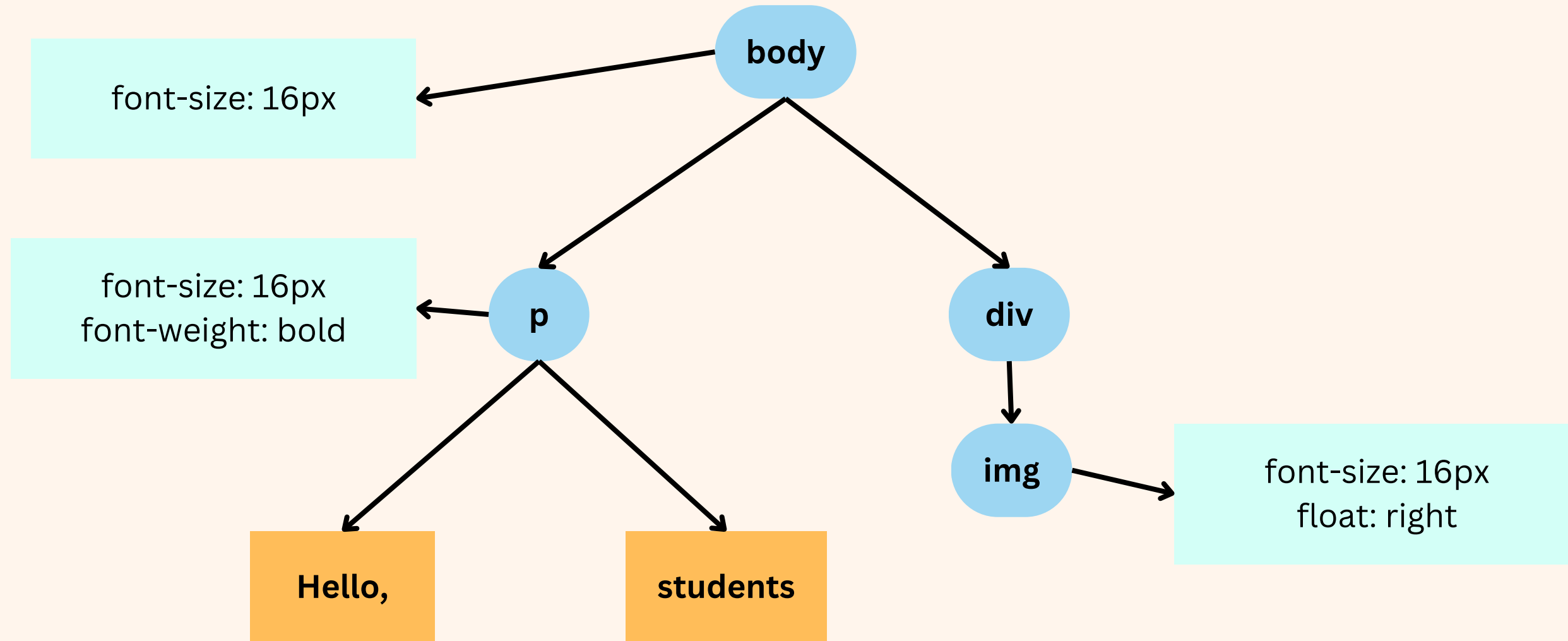
4. CREATE RENDER TREE



4. CREATE RENDER TREE

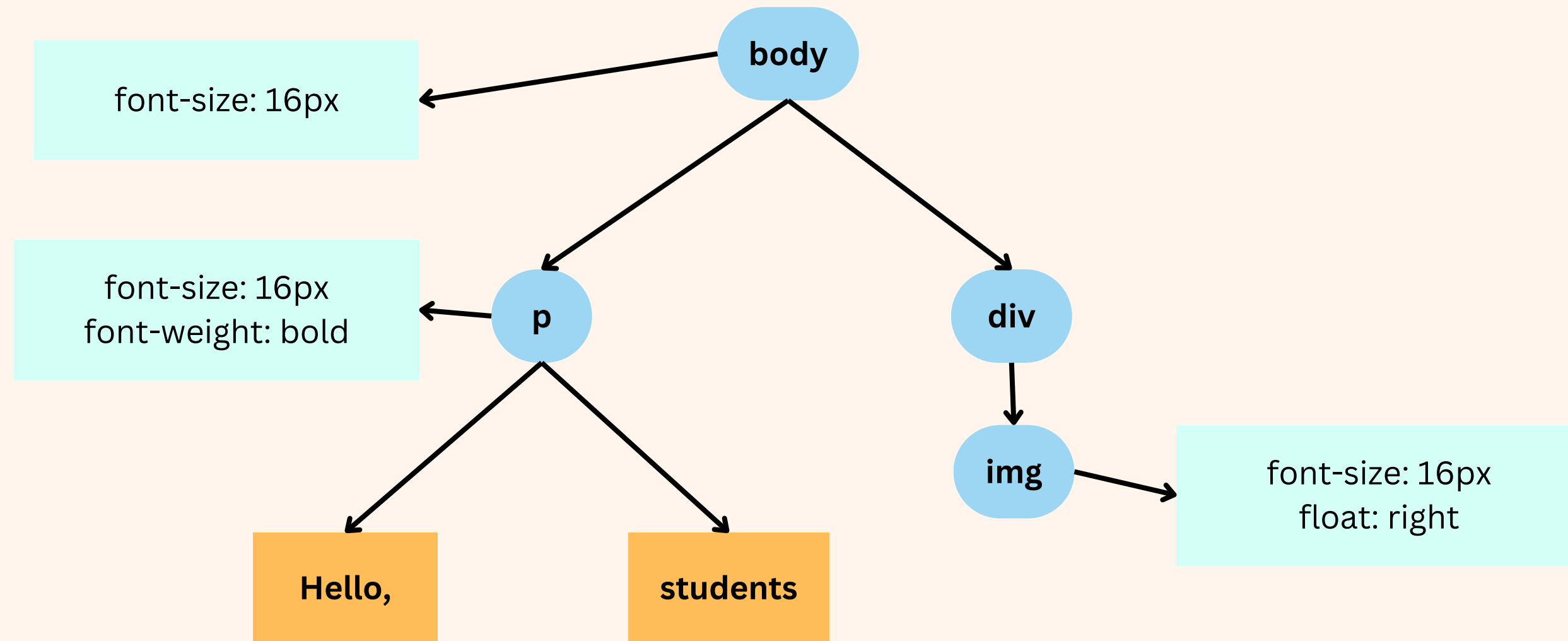


4. CREATE RENDER TREE



Resultant Render Tree

4. CREATE RENDER TREE



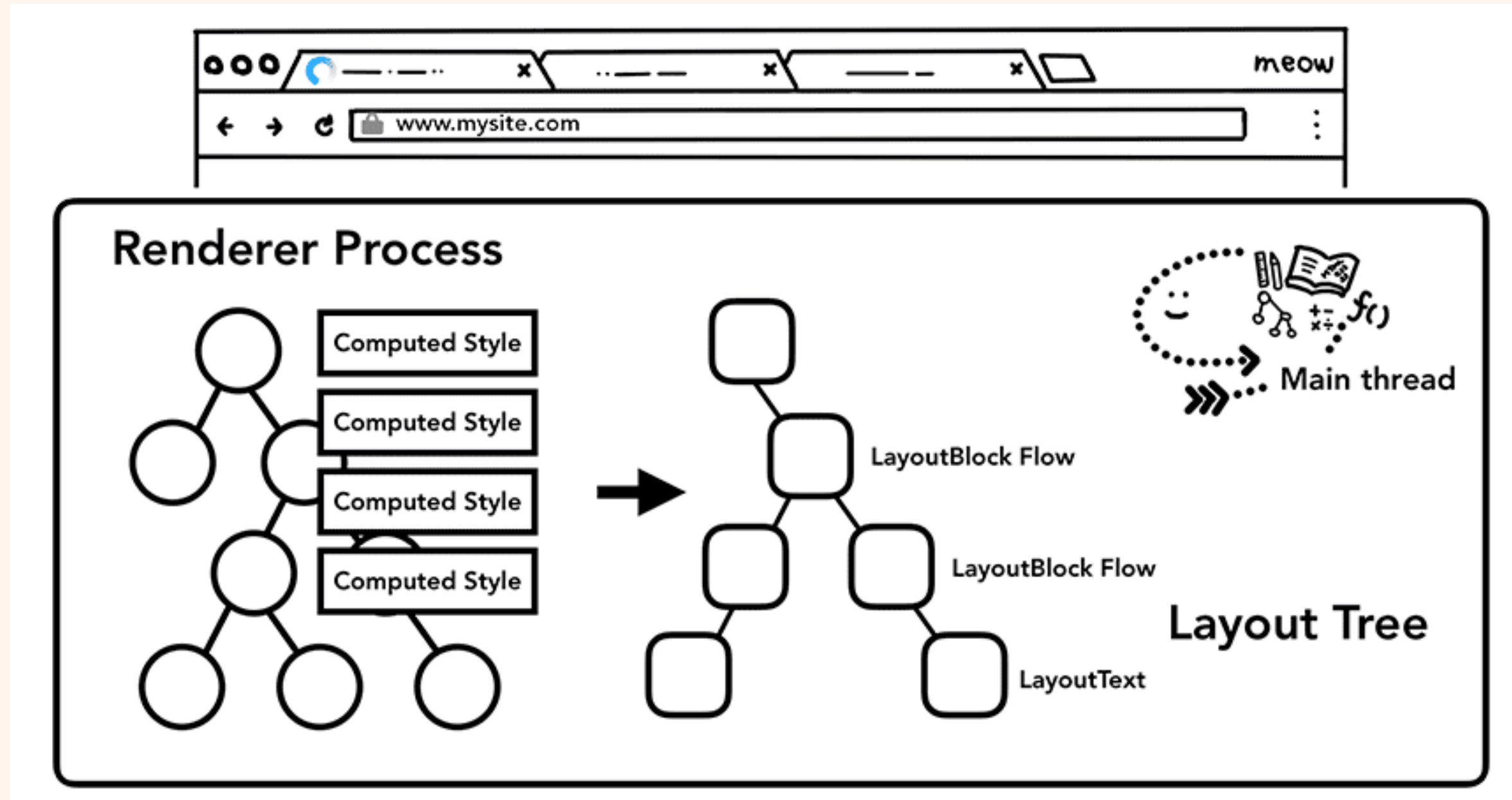
Resultant Render Tree

CRITICAL RENDERING PATH

1. Get the initial HTML
2. Parse the HTML into the DOM
3. Parse CSS into the CSSOM
4. Create render tree
5. Determine where render tree nodes should appear on the page (layout)
6. Paint nodes onto screen (rasterize)

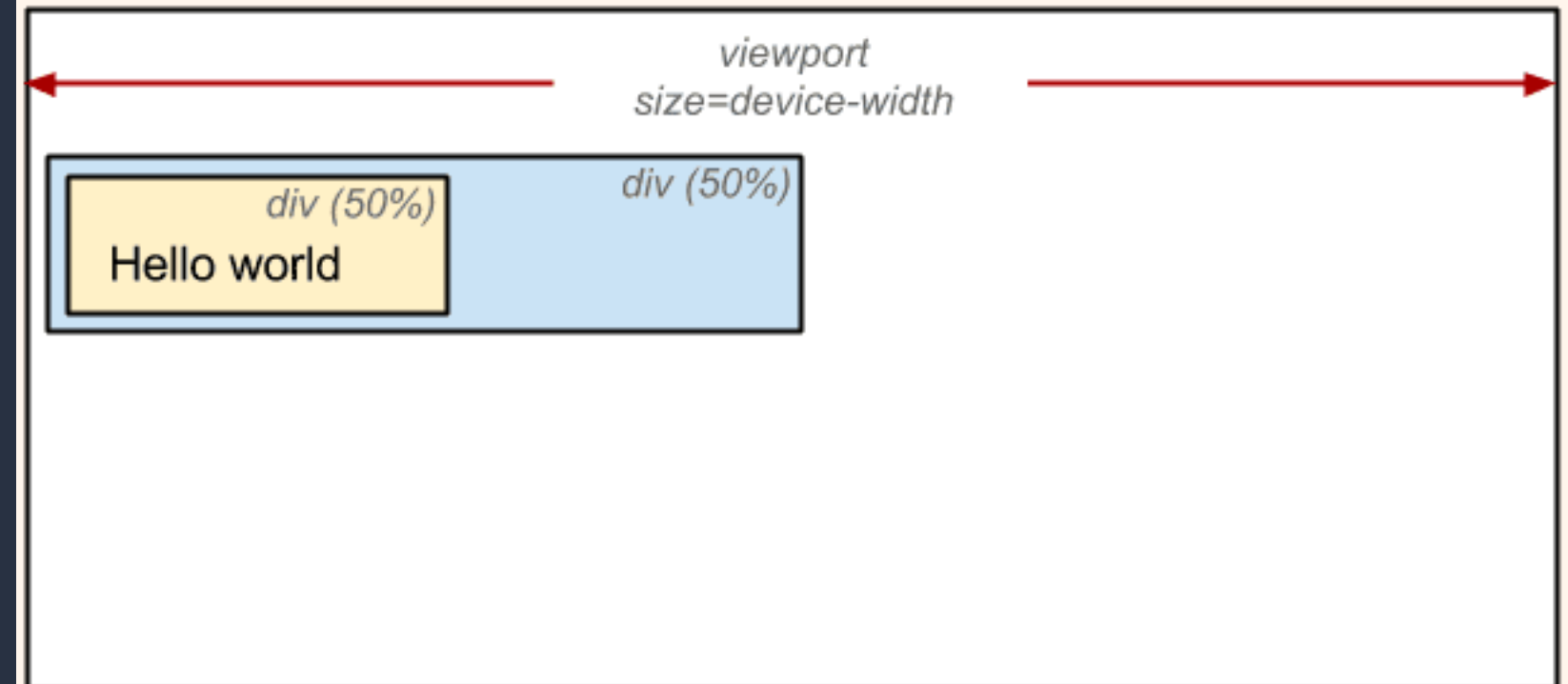


5. LAYOUT



5. LAYOUT

```
<!DOCTYPE html>
<html>
  <head>
    <meta name="viewport" content="width=device-width,initial-scale=1" />
    <title>Critical Path: Hello world!</title>
  </head>
  <body>
    <div style="width: 50%">
      <div style="width: 50%">Hello world!</div>
    </div>
  </body>
</html>
```



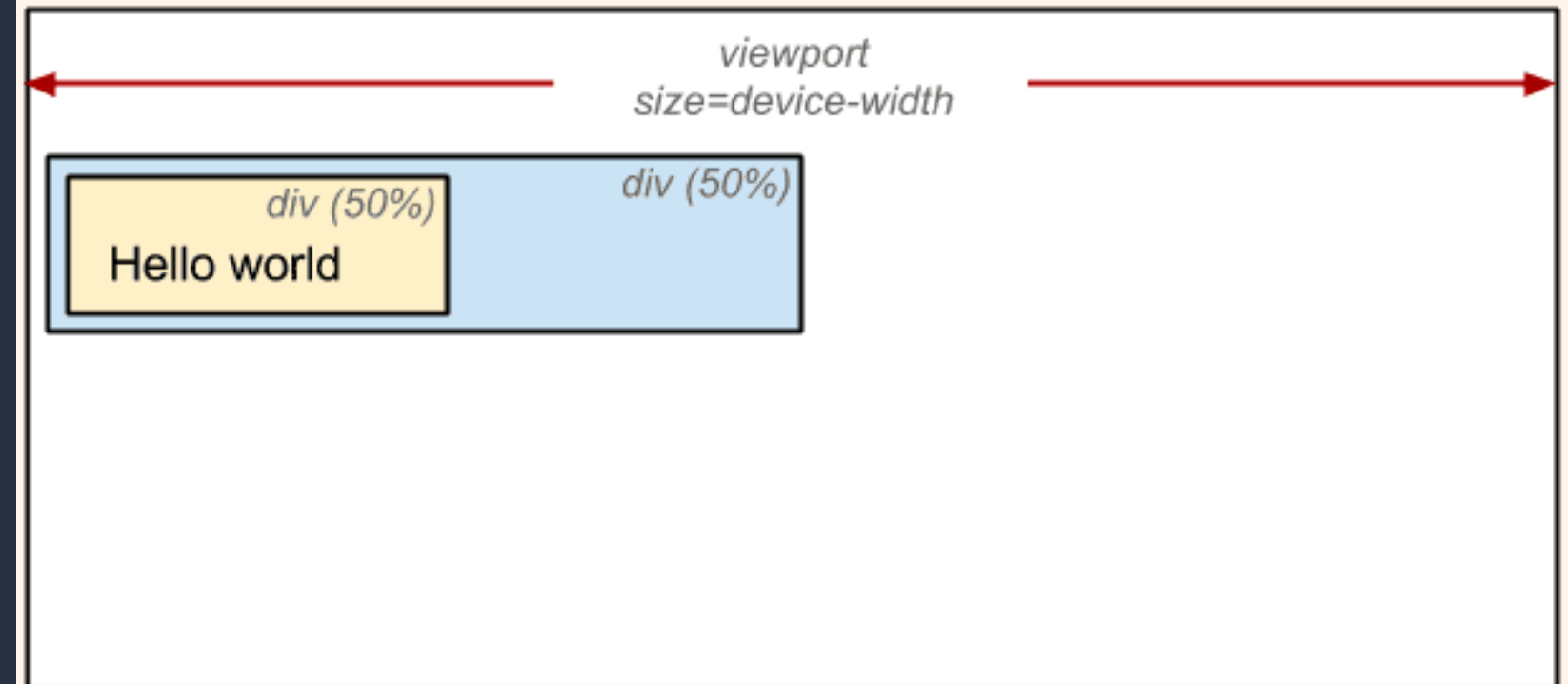
CRITICAL RENDERING PATH

1. Get the initial HTML
2. Parse the HTML into the DOM
3. Parse CSS into the CSSOM
4. Create render tree
5. Determine where render tree nodes should appear on the page (layout)
6. Paint nodes onto screen (rasterize)

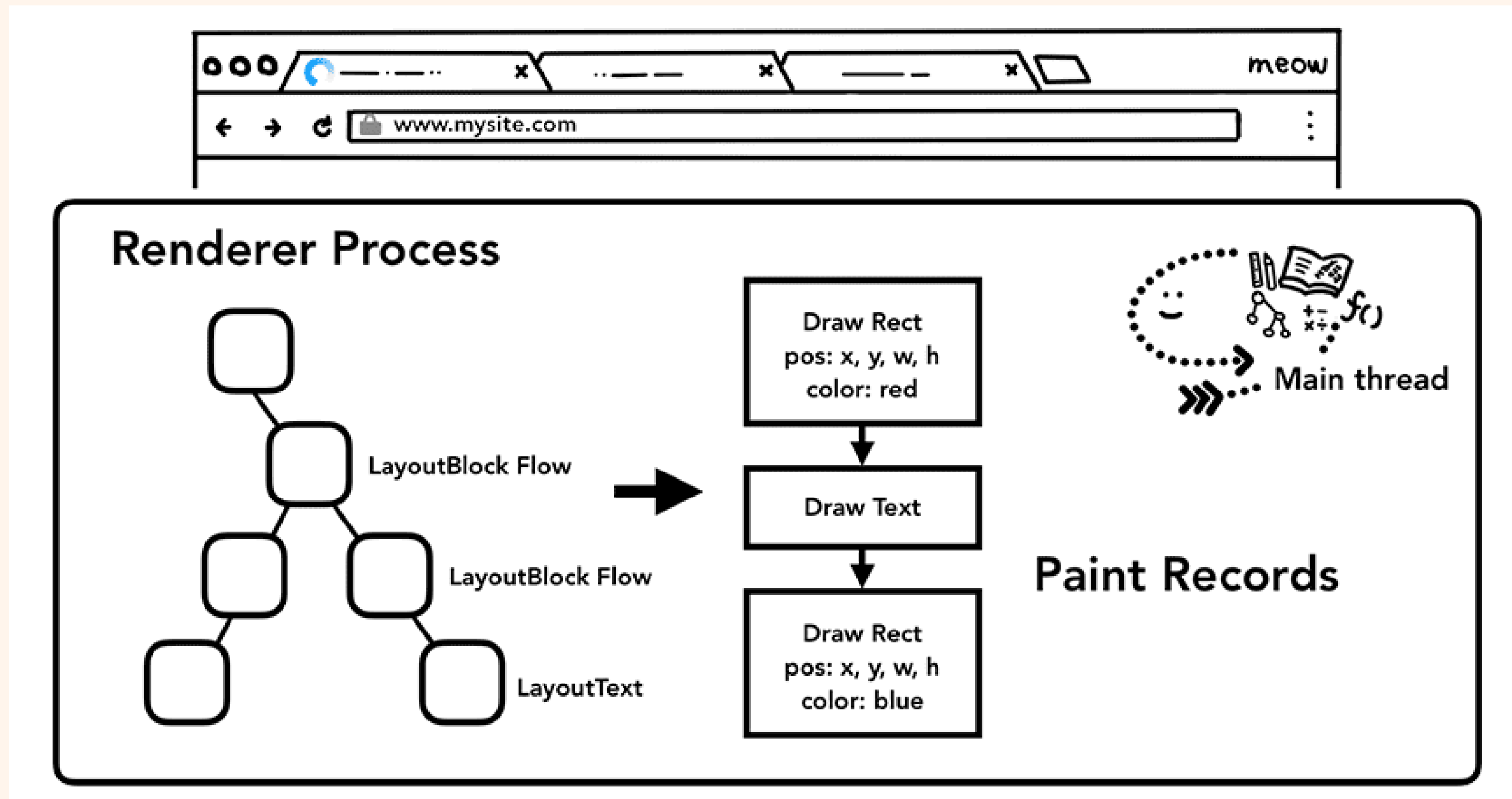


5. LAYOUT

```
<!DOCTYPE html>
<html>
  <head>
    <meta name="viewport" content="width=device-width,initial-scale=1" />
    <title>Critical Path: Hello world!</title>
  </head>
  <body>
    <div style="width: 50%">
      <div style="width: 50%">Hello world!</div>
    </div>
  </body>
</html>
```



6. RASTERIZE/PAINT



6. RASTERIZE/PAINT

Hello students



NEXT TIME....

FE Frameworks Deep Dive